

Chapter II

Metric Spaces; Equivalent Spaces; Classification of Subsets; and the Space of Fractals

1 Spaces

In fractal geometry we are concerned with the structure of subsets of various very simple “geometrical” spaces. Such a space is denoted by X . It is the space on which we think of drawing our fractals; it is the place where fractals live. What is a fractal? For us, for now, it is just a subset of a space. Whereas the space is simple, the fractal subset may be geometrically complicated.

Definition 1.1 *A space X is a set. The points of the space are the elements of the set.*

Although this definition does not say it, the nomenclature “space” implies that there is some structure to the set, some sense of which points are close to which. We give some examples to show the sort of thing this may mean. Throughout this text $\mathbb{R}$ denotes the set of real numbers, and “ $\in$ ” means “belongs to.”

Examples

1.1. $X = \mathbb{R}$. Each “point” $x \in X$ is a real number, or a dot on a line.

1.2. $X = C[0, 1]$, the set of continuous functions that take the real closed interval $[0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$ into the real line $\mathbb{R}$. A “point” $f \in X$ is a function $f : [0, 1] \xrightarrow{cts} \mathbb{R}$. f may be represented by its graph.

Figure II.1. A point x in $\mathbb{R}$.

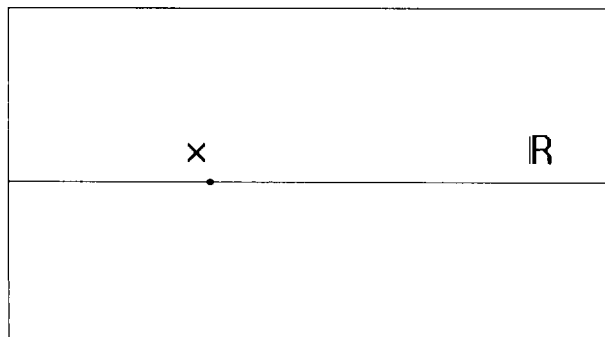
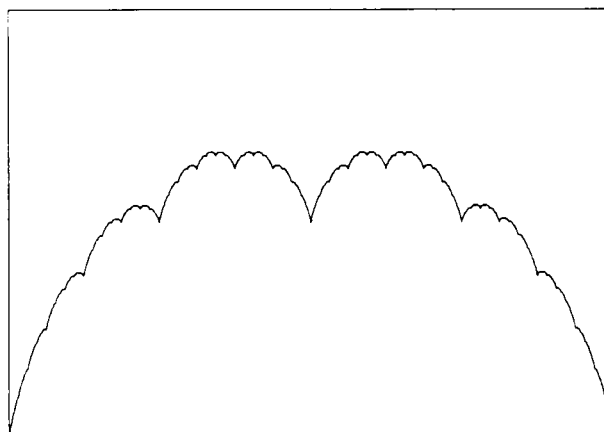


Figure II.2. A point f in the space of continuous functions on $[0, 1]$.



Notice that here $f \in \mathbf{X}$ is not a point on the x -axis; it is the whole function. A continuous function on an interval is characterized by the fact that its graph is unbroken; as a picture it contains no rips or tears; it can be drawn without removing the pencil from the paper.

1.3. $\mathbf{X} = \mathbb{R}^2$, the Euclidean plane, the coordinate plane of calculus. Any pair of real numbers $x_1, x_2 \in \mathbb{R}$ determines a single point in $\mathbb{R}^2$. A point $x \in \mathbf{X}$ is represented in several equivalent ways:

$$x = (x_1, x_2) = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \text{a point in a figure such as Figure II.3.}$$

The spaces in examples 1.1, 1.2, and 1.3 are each *linear spaces*: there is an obviously defined way, in each case, of adding two points in the space to obtain a new one in the same space. In 1.1 if x and $y \in \mathbb{R}$, then $x + y$ is also in $\mathbb{R}$; in 1.2 we define $(f + g)(x) = f(x) + g(x)$; and in 1.3 we define

$$x + y = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix}.$$

Similarly, in each of the above examples, we can multiply members of $\mathbf{X}$ by a scalar, that is, by a real number $\alpha \in \mathbb{R}$. For example, in 1.2 $(\alpha f)(x) = \alpha f(x)$ for any $\alpha \in \mathbb{R}$, and $\alpha f \in C[0, 1]$ whenever $f \in C[0, 1]$. Example 1.1 is a one-dimensional linear space; 1.2 is an ∞ -dimensional linear space (can you think why the dimension is infinite?); and 1.3 is a two-dimensional linear space. A linear space is also called a vector space. The scalars may be complex numbers instead of real numbers.

1.4. The complex plane, $\mathbf{X} = \mathbb{C}$, where any point $x \in \mathbf{X}$ is represented

$$x = x_1 + ix_2, \quad \text{where } i = \sqrt{-1},$$

for some pair of real numbers $x_1, x_2 \in \mathbb{R}$. Any pair of numbers $x_1, x_2 \in \mathbb{R}$ determines a point of $\mathbb{C}$. It is obvious that $\mathbb{C}$ is essentially the same as $\mathbb{R}^2$, but there is an implied distinction. In $\mathbb{C}$ we can multiply two points x, y and obtain a new point in $\mathbb{C}$. Specifically, we define

$$x \cdot y = (x_1 + ix_2)(y_1 + iy_2) = (x_1y_1 - x_2y_2) + i(x_2y_1 + x_1y_2)$$

1.5. $\mathbf{X} = \hat{\mathbb{C}}$, the Riemann sphere. Formally, $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$; that is, all the points of $\mathbb{C}$ together with the “point at infinity.” Here is a way of constructing and thinking about $\hat{\mathbb{C}}$. Place a sphere on the plane $\mathbb{C}$, with the South Pole on the origin, and the North Pole N vertically above it.

To a given point $x \in \mathbb{C}$ we associate a point x' on the sphere by constructing the straight line from N to x and marking where this line intersects the sphere. This associates a unique point $x' = h(x)$ with each point $x \in \mathbb{C}$. The transformation $h : \mathbb{C} \rightarrow \text{sphere}$ is clearly continuous in the sense that nearby points go to nearby points. Points farther and farther away from 0 in the plane $\mathbb{C}$ end up closer and closer to N . $\hat{\mathbb{C}}$ consists of the completion of the range of h by including N on the sphere: The “point at infinity (∞)” can be thought of as a giant circle, infinitely far out in $\mathbb{C}$, whose image under h is N . It is easier to think of $\hat{\mathbb{C}}$ being the whole of the sphere, rather than as the plane together with ∞ . It is of interest that $h : \mathbb{C} \rightarrow \text{sphere}$

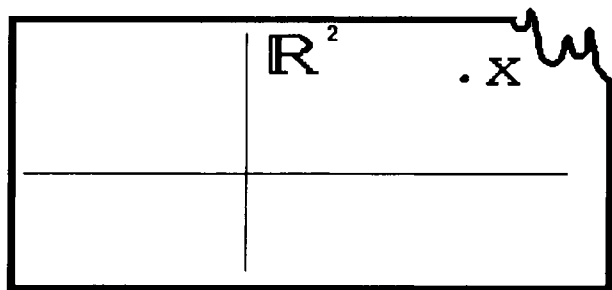
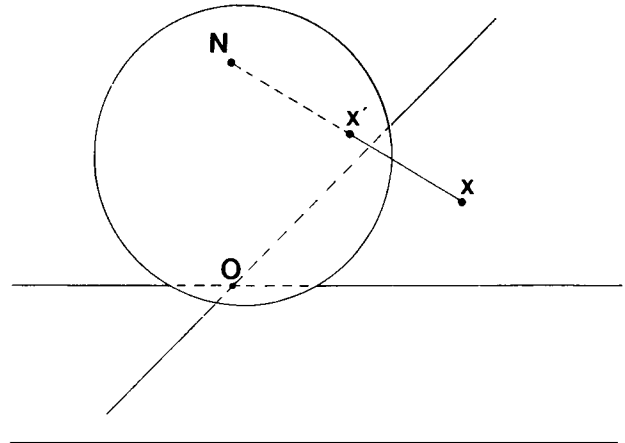


Figure 11.3. A point x in the space $\mathbb{R}^2$.

Figure II.4. Construction of a geometrical representation for the Riemann sphere. N is the North Pole and corresponds to the “point at infinity.”



is *conformal*: it preserves angles. The image under h of a triangle in the plane is a curvaceous triangle on the sphere.

Although the sides of the triangle on the sphere are curvaceous they meet in well-defined angles, as one can visualize by imagining the globe to be magnified enormously. The angles of the curvaceous triangle are the same as the corresponding angles of the triangle in the plane.

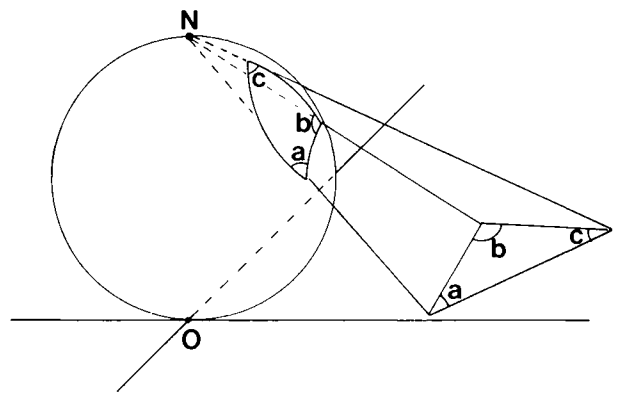
Examples & Exercises

1.6. $\mathbf{X} = \Sigma$, the *code space* on N symbols. Usually the symbols are the integers $\{0, 1, 2, \dots, N - 1\}$. A typical point in $\mathbf{X}$ is a semi-infinite word such as

$$x = 2 \ 17 \ 0 \ 0 \ 1 \ 21 \ 15 \ (N - 1) \ 3 \ 0 \ \dots$$

There are infinitely many symbols in this sequence. In general, for a given element $x \in \mathbf{X}$, we can write

Figure II.5. A triangle in the plane corresponds to a curvaceous triangle on the sphere.



$$x = x_1x_2x_3x_4x_5x_6x_7x_8\ldots, \quad \text{where each } x_i \in \{0, 1, 2, \ldots, N-1\}.$$

There are many names attached to this space because of its importance in a variety of branches of mathematics and physics. When each symbol is intended to represent a random choice from N possibilities, each point in this space represents a particular sequence of events, from a set of N possible events. In this case, the space is sometimes called the space of Bernoulli trials. When there are several code spaces being referred to, it is customary to write the code space on N symbols as Σ_N .

1.7. A few other favorite spaces are defined as follows.

- (a) A disk in the plane with center at the origin and with finite radius $R > 0$:

$$\bullet = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq R^2\}.$$

- (b) A “filled” square:

$$\blacksquare = \{x \in \mathbb{R}^2 : 0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1\}.$$

- (c) An interval:

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \text{ where } a \text{ and } b \text{ are real numbers with } a < b.$$

- (d) Body space:



$$= \{x \in \mathbb{R}^3 : \text{coordinate points implied by a cadaver frozen in } \mathbb{R}^3\}.$$

- (e) Sierpinski space

$$\triangle = \{x \in \mathbb{R}^2 : x \text{ is a point on a certain fixed Sierpinski triangle}\}.$$

Sierpinski triangles occur often in this text. See, for example, Figure IV.94.

1.8. Show that the examples in 1.5, 1.6, and 1.7 are not vector spaces, at least if addition and multiplication by reals are defined in the usual way.

1.9. The notation $A \subset X$ means A is a *subset* of X ; that is, if $x \in A$ then $x \in X$, or $x \in A$ implies $x \in X$. The symbol $\emptyset$ means the empty set. It is defined to be the set such that the statement “ $x \in \emptyset$ ” is always false. We use the notation $\{x\}$ to denote the set consisting of a single point $x \in X$. Show that if $x \in X$, then $\{x\}$ is a subset of X .

1.10. Any set of points makes a space, if we care to define it as such. The points are what we choose them to be. Why, do you think, have the spaces defined above been picked out as important? Describe other spaces that are equally important.

1.11. Let X_1 and X_2 be spaces. These can be used to make a new space denoted $X_1 \times X_2$, called the Cartesian product of X_1 and X_2 . A point in $X_1 \times X_2$ is represented by the ordered pair (x_1, x_2) , where $x_1 \in X_1$ and $x_2 \in X_2$. For example, $\mathbb{R}^2$ is the Cartesian product of $\mathbb{R}$ and $\mathbb{R}$.

1.12. As another example of a Cartesian product let

$$X = \{(x, y) : x, y \in \Sigma\} = \Sigma \times \Sigma,$$

where Σ is the code space on N symbols. This has an interpretation in terms of the random choices mentioned in exercise 1.6. We call y the past and x the future. Then each element of the space represents a sequence of “coin tosses” (the coins are really more like N -sided dice); y represents the tosses that have already happened, beginning with the latest one, and x represents the tosses to come (beginning with the next one). If we rewrite the point (x, y) with a dot marking the “present,”

$$\dots y_3 y_2 y_1 \cdot x_1 x_2 x_3 \dots$$

then the act of moving the dot to the right moves one future coin toss to a past coin toss; the obvious interpretation is that it represents flipping the coin. Moving the dot is called a *shift*, and the space is called *the space of shifts on N symbols*. It is also denoted Σ , whether it is this space or code space that is being referred to is usually clear from context. In this book Σ will always be code space unless specifically mentioned.

2 Metric Spaces

We use the notation “ $\forall$ ” to mean “for all.” We also introduce the notation $A \setminus B$ to mean the set A “take away” the set B . That is, $A \setminus B = \{x \in A : x \notin B\}$. We use “ $\Rightarrow$ ” to mean “implies.”

Definition 2.1 A metric space (X, d) is a space X together with a real-valued function $d : X \times X \rightarrow \mathbb{R}$, which measures the distance between pairs of points x and y in X . We require that d obeys the following axioms:

- (1) $d(x, y) = d(y, x) \forall x, y \in X$
- (2) $0 < d(x, y) < \infty \forall x, y \in X, x \neq y$
- (3) $d(x, x) = 0 \forall x \in X$
- (4) $d(x, y) \leq d(x, z) + d(z, y) \forall x, y, z \in X$.

Such a function d is called a metric.

The concept of shortest paths between points in a space, *geodesics*, is dependent on the metric. The metric may determine a *geodesic structure* of the space. Geodesics on a sphere are great circles; in the plane with the Euclidean metric they are straight lines.

Examples & Exercises

2.1. Show that the following are all metrics in the space $X = \mathbb{R}$:

- (a) $d(x, y) = |x - y|$ (Euclidean metric)
- (b) $d(x, y) = 2 \cdot |x - y|$
- (c) $d(x, y) = |x^3 - y^3|$

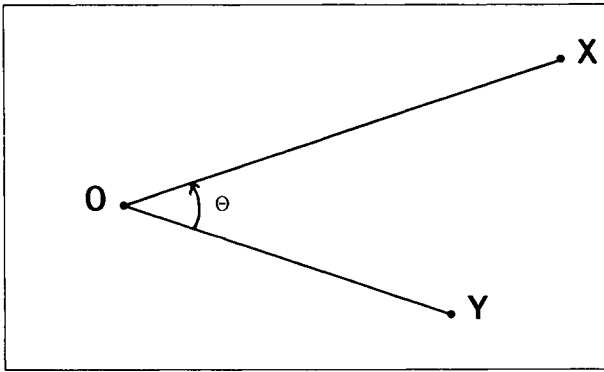


Figure 11.6. (The angle θ , and the distances r_1, r_2 used to construct a metric on the punctured plane.) Acute angle subtended by two straight lines.

2.2. Show that the following are metrics in the space $\mathbf{X} = \mathbb{R}^2$:

- (a) $d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$ (Euclidean metric)
- (b) $d(x, y) = |x_1 - y_1| + |x_2 - y_2|$ (Manhattan metric)

Why is the name “Manhattan” used in connection with (b)?

2.3. Show that $d(x, y) = |xy|$ does not define a metric in $\mathbb{R}$.

2.4. Let $\mathbb{R}^2 \setminus \{O\}$ denote the punctured plane. Define $d(x, y)$ as follows:

$$d(x, y) = |r_1 - r_2| + |\theta|,$$

where r_1 = Euclidean distance from x to O , r_2 = Euclidean distance from y to O , O is the origin, and θ is the smallest angle subtended by the two straight lines connecting x and y to the origin. Show that d is a metric.

2.5. On the code space Σ define

$$d(x, y) = d(x_1x_2x_3\ldots, y_1y_2y_3\ldots) = \sum_{i=1}^{\infty} \frac{|x_i - y_i|}{(N+1)^i}$$

Show that every pair of points in Σ is a *finite* distance apart. That is, d is indeed a function that takes $\Sigma \times \Sigma$ into $\mathbb{R}$. Verify that (Σ, d) is a metric space. Try to envisage a possible geometry for Σ . (Do not confuse the possible meanings of the symbol Σ ; from its context it should be clear when it refers to code space and when it refers to summation.)

***2.6.** Define $\mathbf{X} = \{(x, y) : x, y \in \Sigma\}$, the space of shifts on N symbols as in exercise 1.12. We can define a *Euclidean* distance by treating each coordinate as a base $N+1$ number between 0 and 1. That is, we make the distance between (x, y) and (u, v) equal to

$$\sqrt{\left(\sum_{i=1}^{\infty} \frac{x_i - u_i}{(N+1)^i}\right)^2 + \left(\sum_{i=1}^{\infty} \frac{y_i - v_i}{(N+1)^i}\right)^2}$$

Show that this is indeed a metric space.

2.7. In $X =$  define $d(x, y)$ to be the Euclidean length of the shortest path lying entirely within X which connects x and y . Show that this is a metric. Discuss the utility of this metric in anatomy. The distance from a toenail to a fingertip does not depend much on the configuration of the body, whereas the usual spatial distance would.

2.8. Invent a function $d : \blacksquare \times \blacksquare \rightarrow \mathbb{R}$ which is not a metric. Define a metric for the space , namely an annulus, which makes it seem like the curved wall of a cylinder:

***2.9.** Show that a metric on $X = \hat{\mathbb{C}}$ is defined by shortest great circle distances on the sphere. Compare the distances from 0, and from $1 + i$, to ∞ .

Definition 2.2 Two metrics d_1 and d_2 on a space X are equivalent if there exist constants $0 < c_1 < c_2 < \infty$ such that

$$c_1 d_1(x, y) \leq d_2(x, y) \leq c_2 d_1(x, y), \quad \forall (x, y) \in X \times X.$$

Examples & Exercises

2.10. Definition 2.2 looks unsymmetrical; it does not appear to make the same requirements on d_1 as it does on d_2 . Show that this is an illusion by establishing that if the definition holds then there are constants $0 < e_1 < e_2 < \infty$ so that

$$e_1 d_2(x, y) \leq d_1(x, y) \leq e_2 d_2(x, y), \quad \forall (x, y) \in X \times X.$$

2.11. Are the Manhattan and Euclidean metrics equivalent on $\blacksquare \subset \mathbb{R}^2$? What about on $\mathbb{R}^2$?

2.12. Show that the metric in exercise 2.4 is *not* equivalent to the Euclidean metric on  $\setminus \{0\}$.

One notion underlying the concept of equivalent metrics is that any pair of equivalent metrics gives the same notion of which points are close together and which are far apart. It is as though there were a standard way for boundedly deforming the space, whereby distances are determined both before and after deformation.

For example, consider a pair of points x and y in $\blacksquare \subset \mathbb{R}^2$. Let the Euclidean distance between these points be $d_1(x, y)$. Think of a thin rubber sheet lying over $\blacksquare$. This sheet is stretched in some repeatable fashion, carrying copies of the points x and y to new locations, as illustrated in Figure II.7. The Euclidean distance between these moved points is called $d_2(x, y)$. The condition of equivalence is the requirement that there is no extreme (infinite) stretching or compression of the space.

This leads us to the idea of equivalent metric spaces:

Definition 2.3 Two metric spaces (X_1, d_1) and (X_2, d_2) are equivalent if there is a function $h: X_1 \rightarrow X_2$ that is one-to-one and onto (i.e., it is invertible), such that

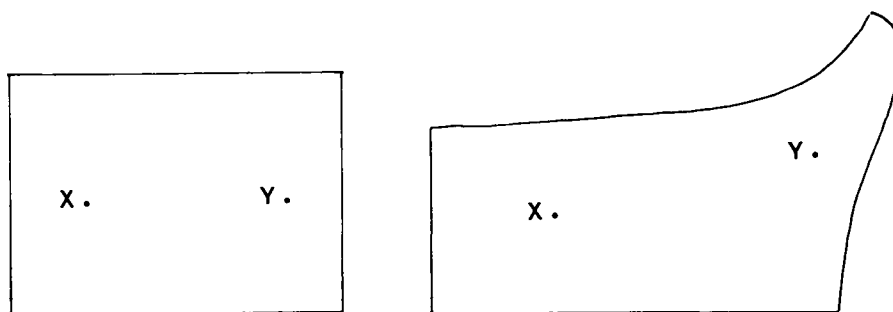


Figure II.7. A thin rubber sheet lies over the $\blacksquare$ in the plane and is stretched. The Euclidean distances between points are determined before and after deformation, yielding two metrics. These metrics may be equivalent if the deformation leads to no rips, tears, or infinite stretching.

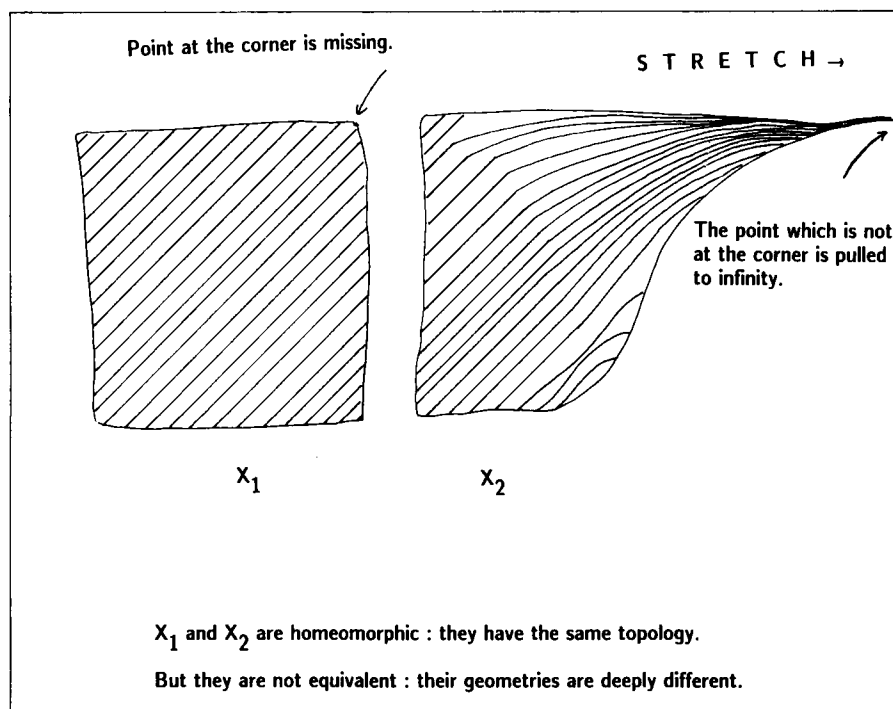


Figure II.8. This picture suggests two metric spaces X_1 and X_2 that have the same topology, but that are not metrically equivalent: their “geometries” are deeply different.

the metric $\tilde{d}_1$ on X_1 defined by

$$\tilde{d}_1(x, y) = d_2(h(x), h(y)), \quad \forall x, y \in X_1$$

is equivalent to d_1 .

One can think of Definition 2.3 as requiring that X_1 and X_2 are related to one another by a bounded deformation, and nowhere is there an arbitrarily large compression or stretching; also, there is no overlapping, folding, or ripping.

Definition 2.4 A function $f : X_1 \rightarrow X_2$ from a metric space (X_1, d_1) into a metric space (X_2, d_2) is continuous if, for each $\epsilon > 0$ and $x \in X_1$, there is a $\delta > 0$ so that

$$d_1(x, y) < \delta \Rightarrow d_2(f(x), f(y)) < \epsilon.$$

If f is also one-to-one and onto, and thus invertible, and if also the inverse f^{-1} of f is continuous, then we say that f is a homeomorphism between X_1 and X_2 . In such a case we say that X_1 and X_2 are homeomorphic.

The assertion that two spaces are equivalent metrically is much stronger than the statement that they are homeomorphic: to be equivalent there must be a bounded relationship between ϵ and δ independent of x . Homeomorphism is the equivalence relationship for topological properties; two spaces that are homeomorphic are identical *topological* spaces. Two metrics d_1 and d_2 on a given space X are identical topologically (define the same topological space) if the identity map $\iota : (X, d_1) \rightarrow (X, d_2)$ given by $\iota(x) = x$ is a homeomorphism.

Examples & Exercises

2.13. Let $X_1 = [1, 2]$ and $X_2 = [0, 1]$. Let d_1 denote the Euclidean and let $d_2(x, y) = 2 \cdot |x - y|$ in X_2 . Show that (X_1, d_1) and (X_2, d_2) are equivalent metric spaces.

2.14. Show that $(\blacksquare, \text{Euclidean})$ and $(\blacksquare, \text{Manhattan})$ are equivalent metric spaces.

2.15. Show that $(\mathbb{C}, \text{Euclidean})$ and $(\mathbb{R}^2, \text{Manhattan})$ are equivalent metric spaces.

2.16. Define two different metrics on the space $X = (0, 1] = \{x \in \mathbb{R} : 0 < x \leq 1\}$ by

$$d_1(x, y) = |x - y| \quad \text{and} \quad d_2(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|.$$

Show that (X, d_1) and (X, d_2) are not equivalent metric spaces.

2.17. Figure II.9 suggests a subset (black) of $(\blacksquare, \text{Euclidean})$. It also shows the space and set deformed by a metric equivalence. Discuss the properties of the image that would be invariant under (a) any metric equivalence, and (b) any homeomorphism. To what extent might one be able to “see” these invariances? Think about how much deformation an image can withstand while remaining recognizably the same image. Look at reflections of sets and images in the back of a shiny spoon.

2.18. Show that if two metric spaces are metrically equivalent then there is a homeomorphism between them.

***2.19.** We can define metrics on Σ , our code space, by

$$d_k(x, y) = \sum_{i=1}^{\infty} \frac{|x_i - y_i|}{k^i},$$

where $k > 1$. The most important choices for k are $N + 1$ and N , but other choices (mostly real numbers between these two) are useful. Show that the functions $d_k : \Sigma \times \Sigma \rightarrow \mathbb{R}$ are metrics, and that these metrics are identical topologically.

3 Cauchy Sequences, Limit Points, Closed Sets, Perfect Sets, and Complete Metric Spaces

Fractal geometry is concerned with the description, classification, analysis and observation of subsets of metric spaces (X, d) . The metric spaces are usually, but not always, of an inherently “simple” geometrical character; the subsets are typically geometrically “complicated.” There are a number of general properties of subsets of metric spaces, which occur over and over again, which are very basic, and which form part of the vocabulary for describing fractal sets and other subsets of metric spaces. Some of these properties, such as openness and closedness, which we are going to introduce, are of a *topological* character. That is to say, they are invariant under homeomorphism.

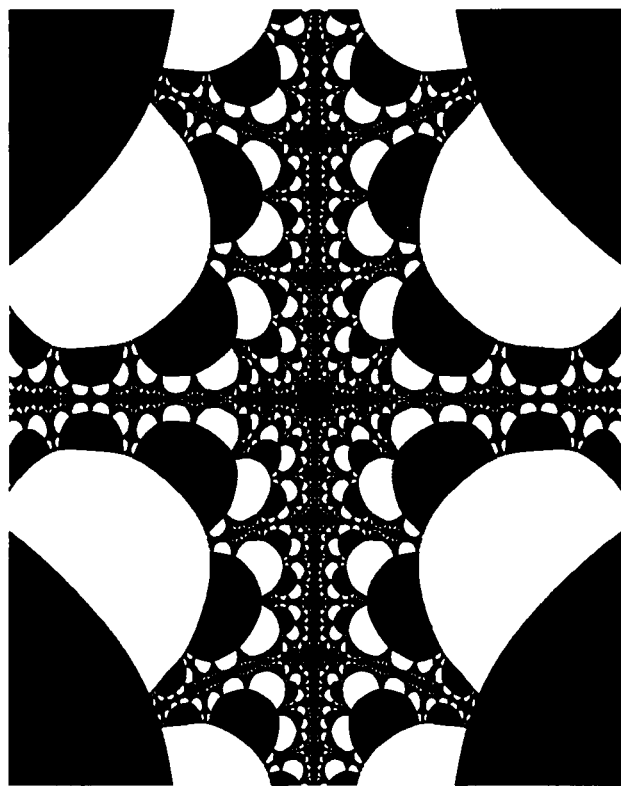


Figure 11.9. What features of the set (black) are invariant under a metric equivalence transformation? Two sets that are metrically equivalent to (a) are shown in (b) and (c).

Figure II.9. (b)

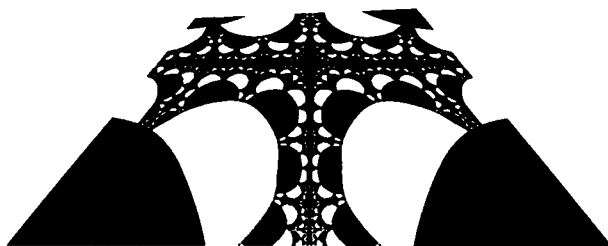
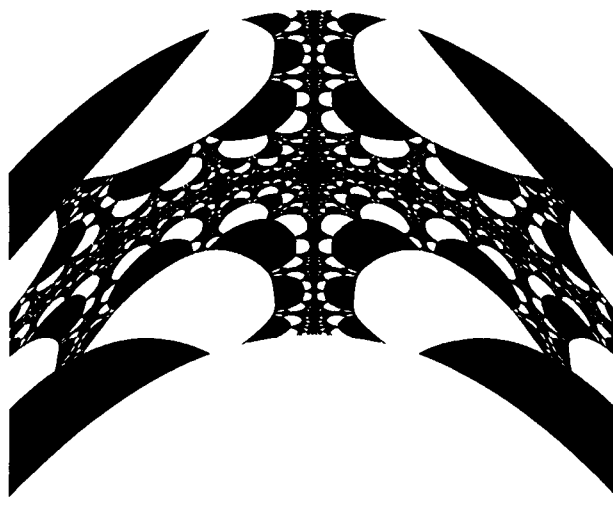


Figure II.9. (c)



For us what is important, however, is that there is another class of properties that are invariant under metric space equivalence. These include openness, closedness, boundedness, completeness, compactness, and perfection; these properties are introduced in this and the next section. Later we will discover another such property, the fractal dimension of a set. If a subset of a metric space has one of these properties, and the space is deformed with bounded distortion, then the corresponding subsets in the deformed space still have that same property.

We are also about another business in this section. In our search for fractals we are always going to look in a certain type of metric space known as “complete.” We need to understand this concept.

Definition 3.1 A sequence $\{x_n\}_{n=1}^{\infty}$ of points in a metric space (X, d) is called a Cauchy sequence if, for any given number $\epsilon > 0$, there is an integer $N > 0$ so that

$$d(x_n, x_m) < \epsilon \quad \text{for all } n, m > N.$$

In other words, the further along the sequence one goes, the closer together become the points in the sequence. Mentally one pictures something like the image in Figure II.10.

However, just because a sequence of points moves closer together as one goes

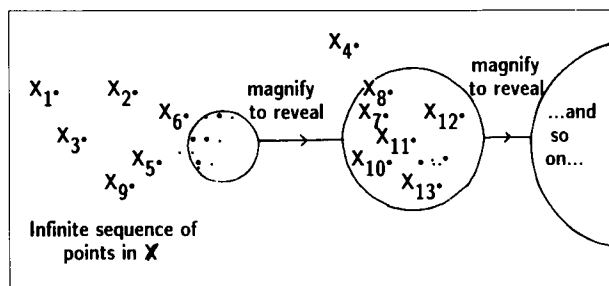


Figure II.10. Image representing successive magnifications on a Cauchy sequence, an infinite sequence of points in X . Just because the points are getting closer and closer together as one looks in at higher magnification does not mean that there is a point x to which the sequence is converging!

along the sequence, we must not infer that they are approaching a point. Perhaps they are trying to approach a point that is not there?

Definition 3.2 A sequence $\{x_n\}_{n=1}^{\infty}$ of points in a metric space (X, d) is said to converge to a point $x \in X$ if, for any given number $\epsilon > 0$, there is an integer $N > 0$ so that

$$d(x_n, x) < \epsilon \quad \text{for all } n > N.$$

In this case the point $x \in X$, to which the sequence converges, is called the limit of the sequence, and we use the notation

$$x = \lim_{n \rightarrow \infty} x_n.$$

The limit x of a convergent sequence $\{x_n\}_{n=1}^{\infty}$ has this property: let

$$B(x, \epsilon) = \{y \in X : d(x, y) \leq \epsilon\}$$

denote a closed ball of radius $\epsilon > 0$ centered at x , as illustrated in Figure II.11.

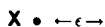
Any such ball centered at x contains all of the points x_n after some index N , where N typically becomes larger and larger as ϵ becomes smaller and smaller. See Figure II.12.

Theorem 3.1 If a sequence of points $\{x_n\}_{n=1}^{\infty}$ in a metric space (X, d) converges to a point $x \in X$, then $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence.

Figure II.11. Uncolored small ball $B(x, \epsilon)$ with its center at x and radius ϵ . Beware! Balls do not usually look like balls. It depends on the metric and on the space. Balls (a)–(c) represent balls (marked in black) in spaces X that are subsets of $\mathbb{R}^2$, with the Euclidean metric. In (a) X has a ragged boundary, viewed as a subset of $\mathbb{R}^2$. In (b) the point $x \in X$ is isolated. In (c) X is a curvaceous Sierpinski triangle. The ball depicted in (d) is in $\mathbb{R}^2$, but the metric is $d(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$.



(a)



(b)

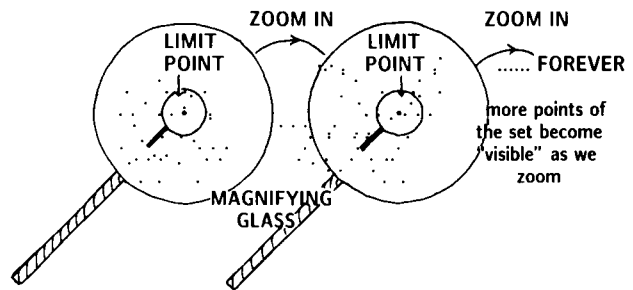


(c)



(d)

Figure II.12. Magnifying glass looking at a magnifying glass near a limit point.



Definition 3.3 A metric space (X, d) is complete if every Cauchy sequence $\{x_n\}_{n=1}^{\infty}$ in X has a limit $x \in X$.

In other words, there actually exists, in the space, a point x to which the Cauchy sequence is converging. This point x is of course the limit of the sequence. If $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence of points in X and if X is complete, then there is a point $x \in X$ such that, for each $\epsilon > 0$, $B(x, \epsilon)$ contains x_n for infinitely many integers n .

We will sometimes use the notation $\{x_n\}$ in place of $\{x_n\}_{n=1}^{\infty}$ and $\lim$ in place of $\lim_{n \rightarrow \infty}$ when it is clear from the context what the domain of the index is.

Examples & Exercises

3.1. Prove that if $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence of points in X and if X is complete, then there is a point $x \in X$ such that, for each $\epsilon > 0$, $B(x, \epsilon)$ contains x_n for infinitely many integers n .

***3.2.** Show that $(\mathbb{R}, \text{Euclidean metric})$ is a complete metric space.

3.3. Show that $(\mathbb{R}^2, \text{Euclidean metric})$ is a complete metric space.

3.4. Show that $(\mathbb{N}, \text{Euclidean metric})$ is a complete metric space.

3.5. Show that $(\hat{\mathbb{C}}, \text{metric on sphere})$ is a complete metric space.

3.6. Show that $(\Sigma, \text{code space})$ is a complete metric space.

3.7. $(C[0, 1], D)$ is a complete metric space, where the metric D is defined by

$$D(f, g) = \max\{|f(s) - g(s)| : s \in [0, 1]\}.$$

3.8. Let (X_1, d_1) and (X_2, d_2) be equivalent metric spaces. Suppose (X_1, d_1) is complete. Show that (X_2, d_2) is complete.

3.9. Show that there are many different “shortest paths” between most pairs of points in $(\mathbb{N}, \text{Manhattan})$.

3.10. Prove Theorem 3.1.

3.11. Prove that any sequence in a metric space can have at most one limit.

Definition 3.4 Let $S \subset X$ be a subset of a metric space (X, d) . A point $x \in X$ is called a limit point of S if there is a sequence $\{x_n\}_{n=1}^{\infty}$ of points $x_n \in S \setminus \{x\}$ such that $\lim_{n \rightarrow \infty} x_n = x$.

Definition 3.5 Let $S \subset X$ be a subset of a metric space (X, d) . The closure of S , denoted $\bar{S}$, is defined to be $\bar{S} = S \cup \{\text{Limit points of } S\}$. S is closed if it contains all of its limit points, that is, $S = \bar{S}$. S is perfect if it is equal to the set of all its limit points.

Exercises & Examples

3.12. Show that 0 is a limit of the sequence $\{x_n = \frac{1}{n}\}_{n=1}^{\infty}$ in the metric space $([0, 1], \text{Euclidean})$ but not in the metric space $((0, 1], \text{Euclidean})$.

3.13. A metric space (X, d) consists of a single point $X = \{a\}$, together with a metric defined by $d(a, a) = 0$. Show that X contains a Cauchy sequence and the limit of the Cauchy sequence, but that it possesses no limit points. Hence show that X is closed and complete but not perfect.

3.14. Show that the sequence $\{x_n = n\}_{n=1}^{\infty}$ has no limit in $(\mathbb{R}, \text{Euclidean})$, but that it does when the points are treated as belonging to $(\hat{\mathbb{C}}, \text{spherical})$.

3.15. Show that if $h : X_1 \rightarrow X_2$ makes the metric spaces (X_1, d_1) and (X_2, d_2) metrically equivalent, then the statements “ $x \in X_1$ is a limit point of $S \subset X_1$ ” and “ $h(x) \in X_2$ is a limit point of $h(S) \subset X_2$ ” are equivalent. Here we use the notation $h(S) = \{h(s) : s \in S\}$.

3.16. Find all of the limit points of the set $\{x_n = (\frac{1}{n} + (-1)^n, \frac{1}{n} + (-1)^{2n}) : n = 1, 2, 3, \dots\}$ in the metric space $(\mathbb{R}^2, \text{Euclidean})$.

3.17. Show that the subset $S = \{x = \frac{1}{n} : n = 1, 2, 3, \dots\}$ is closed in $((0, 1], \text{Euclidean})$.

3.18. Show that $S = [0, 1]$ is a perfect subset of $(\mathbb{R}, \text{Euclidean})$.

3.19. Show that $S = \{\frac{1}{n} : n = 1, 2, 3, \dots\} \cup \{0\}$ is not a perfect subset of $(\mathbb{R}, \text{Euclidean})$, but that $S = \bar{S}$.

3.20. Show that $S = \Sigma$ is a perfect subset of $(\Sigma, \text{code space metric})$.

3.21. Let S be a subset of a complete metric space (X, d) . Then (S, d) is a metric space. Show that (S, d) is complete if, and only if, S is closed in X .

4 Compact Sets, Bounded Sets, Open Sets, and Boundaries

We continue the description of basic properties that are to be used to describe sets and subsets of metric spaces. Where are the fractals? What are they? They are everywhere and soon you will be able to see them. Not just the pictures, which are

shadows of fractals, but in your mind's eye you will find what and where they really are.

Definition 4.1 Let $S \subset \mathbf{X}$ be a subset of a metric space $(\mathbf{X}, d)$. S is compact if every infinite sequence $\{x_n\}_{n=1}^{\infty}$ in S contains a subsequence having a limit in S .

Definition 4.2 Let $S \subset \mathbf{X}$ be a subset of a metric space $(\mathbf{X}, d)$. S is bounded if there is a point $a \in \mathbf{X}$ and a number $R > 0$ so that

$$d(a, x) < R \forall x \in \mathbf{X}.$$

Definition 4.3 Let $S \subset \mathbf{X}$ be a subset of a metric space $(\mathbf{X}, d)$. S is totally bounded if, for each $\epsilon > 0$, there is a finite set of points $\{y_1, y_2, \dots, y_n\} \subset S$ such that whenever $x \in \mathbf{X}$, $d(x, y_i) < \epsilon$ for some $y_i \in \{y_1, y_2, \dots, y_n\}$. This set of points $\{y_1, y_2, \dots, y_n\}$ is called an ϵ -net.

Theorem 4.1 Let $(\mathbf{X}, d)$ be a complete metric space. Let $S \subset \mathbf{X}$. Then S is compact if and only if it is closed and totally bounded.

Proof Suppose that S is closed and totally bounded. Let $\{x_i \in S\}$ be an infinite sequence of points in S . Since S is totally bounded we can find a finite collection of closed balls of radius 1 such that S is contained in the union of these balls. By the Pigeon-Hole Principle (a huge number of pigeons laying eggs in two letter boxes $\Rightarrow$ at least one letter box contains a huge number of angry pigeons), one of the balls, say B_1 , contains infinitely many of the points x_n . Choose N_1 so that $x_{N_1} \in B_1$. It is easy to see that $B_1 \cap S$ is totally bounded. So we can cover $B_1 \cap S$ by a finite set of balls of radius $1/2$. By the Pigeon-Hole Principle, one of the balls, say B_2 , contains infinitely many of the points x_n . Choose N_2 so that $x_{N_2} \in B_2$ and $N_2 > N_1$. We continue in this fashion to construct a *nested* sequence of balls,

$$B_1 \supset B_2 \supset B_3 \supset B_4 \supset B_5 \supset B_6 \supset B_7 \supset B_8 \supset B_9 \supset \dots \supset B_n \supset \dots$$

where B_n has radius $\frac{1}{2^{n-1}}$ and a sequence of integers $\{N_n\}_{n=1}^{\infty}$ such that $x_{N_n} \in B_n$. It is easy to see that $\{x_{N_n}\}_{n=1}^{\infty}$, which is a subsequence of the original sequence $\{x_n\}$, is a Cauchy sequence in S . Since S is closed, and $\mathbf{X}$ is a complete metric space, S is complete as well; see exercise 4.2. So $\{x_n\}$ converges to a point x in S . (Notice that x is exactly $\bigcap_{n=1}^{\infty} B_n$.) Thus, S is compact.

Conversely, suppose that S is compact. Let $\epsilon > 0$. Suppose that there does not exist an ϵ -net for S . Then there is an infinite sequence of points $\{x_n \in S\}$ with $d(x_i, x_j) \geq \epsilon$ for all $i \neq j$. But this sequence must possess a convergent subsequence $\{x_{N_i}\}$. By Theorem 3.1 this sequence is a Cauchy sequence, and so we can find a pair of integers N_i and N_j with $N_i \neq N_j$ so that $d(x_{N_i}, x_{N_j}) < \epsilon$. But $d(x_{N_i}, x_{N_j}) \geq \epsilon$, so we have a contradiction. Thus there *does* exist an ϵ -net. This completes the proof.

Definition 4.4 Let $S \subset \mathbf{X}$ be a subset of a metric space $(\mathbf{X}, d)$. S is open if for each $x \in S$ there is an $\epsilon > 0$ such that $B(x, \epsilon) = \{y \in \mathbf{X} : d(x, y) \leq \epsilon\} \subset S$.

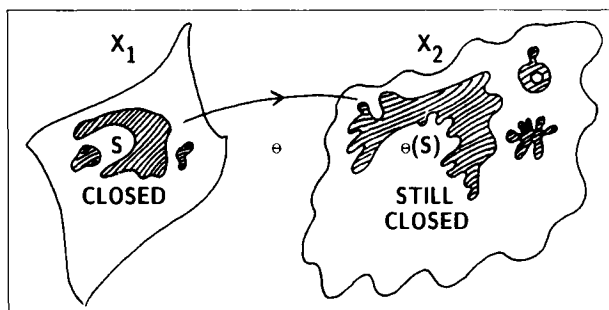


Figure II.13. A transformation θ between two metric spaces, establishing the equivalence of the spaces and carrying the closed set S onto a closed set $\theta(S)$.

Examples & Exercises

4.1. Show that if (X, d) is a metric space then X is closed. Give an example of a metric space that is closed but not complete.

4.2. Let S be a closed subset of a complete metric space (X, d) . Show that (S, d) is a complete metric space.

4.3. Let (X_1, d_1) and (X_2, d_2) be equivalent metric spaces, and let a transformation $\theta : X_1 \rightarrow X_2$ provide this equivalence. Let $S \subset X_1$ be closed. Show that $\theta(S) = \{\theta(s) : s \in S\}$ is closed. This idea is illustrated in Figure II.13.

4.4. If (X, d) is a metric space, then X is open.

Proof Let $x \in X$. Clearly $B(x, 1) \subset X$.

4.5. If (X, d) is a metric space, then “ $S \subset X$ is open” is the same as “ $X \setminus S$ is closed.”

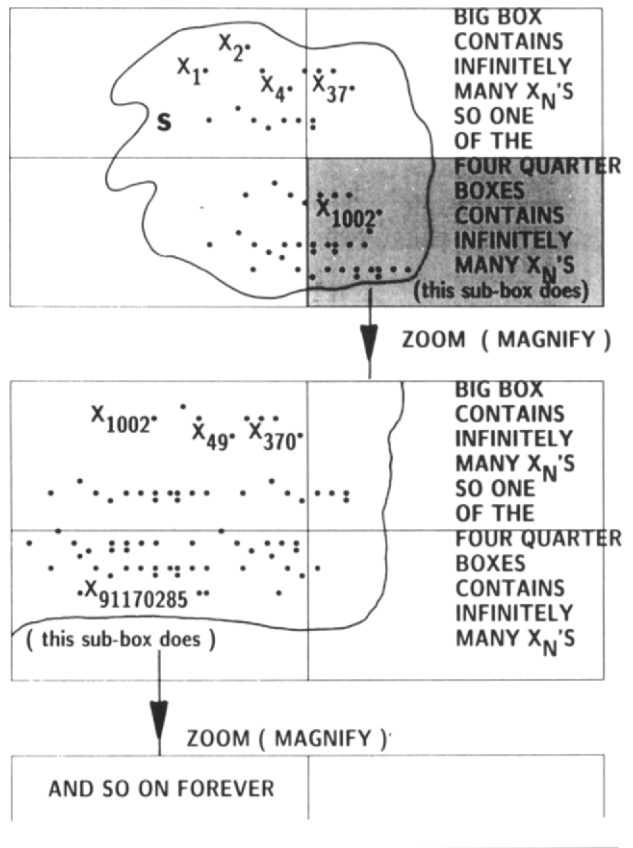
Proof Suppose “ $S \subset X$ is open.” Suppose $\{x_n\}$ is a sequence in $X \setminus S$ with a limit $x \in X$. We must show that $x \in X \setminus S$. Assume that $x \in S$. Then every ball $B(x, \epsilon)$ with $\epsilon > 0$ contains a point $x_n \in X \setminus S$, which means that S is not open. This is a contradiction. The assumption is false. Therefore $x \in X \setminus S$. Therefore “ $X \setminus S$ is closed.”

Suppose “ $X \setminus S$ is closed.” Let $x \in S$. We want to show there is a ball $B(x, \epsilon) \subset S$. Assume there is no ball $B(x, \epsilon) \subset S$. Then for every integer $n = 1, 2, 3, \dots$, we can find a point $x_n \in B(x, \frac{1}{n}) \cap (X \setminus S)$. Clearly $\{x_n\}$ is a sequence in $X \setminus S$, with limit $x \in X$. Since $X \setminus S$ is closed we conclude that $x \in X \setminus S$. This contradicts $x \in S$. The assumption that there is no ball $B(x, \epsilon) \subset S$ is false. Therefore there is a ball $B(x, \epsilon) \subset S$. Therefore “ S is open.”

4.6. Every bounded subset S of $(\mathbb{R}^2, \text{Euclidean})$ has the Bolzano-Weierstrass property: “Every infinite sequence $\{x_n\}_{n=1}^{\infty}$ of points of S contains a subsequence which is a Cauchy sequence.” The proof is suggested by the picture in Figure II.14.

We deduce that every closed bounded subset of $(\mathbb{R}^2, \text{Euclidean})$ is compact. In particular, every metric space of the form (closed bounded subset of $\mathbb{R}^2$, Euclidean)

Figure II.14. Demonstration of the Bolzano-Weierstrass Theorem. (Government warning: This is not a proof.)



is a complete metric space. Show that we can make a rigorous proof by using Theorem 4.1. Begin by proving that any bounded subset of $\mathbb{R}^n$ is totally bounded.

4.7. Let (X, d) be a metric space. Let $f : X \rightarrow X$ be continuous. Let A be a compact nonempty subset of X . Show that $f(A)$ is a compact nonempty subset of X . (This result is proved later as Lemma 7.2 of Chapter III.)

4.8. Let $S \subset (X_1, d_1)$ be open, and let (X_2, d_2) be a metric space equivalent to (X_1, d_1) , the equivalence being provided by a function $h : X_1 \rightarrow X_2$. Show that $h(S)$ is an open subset of X_2 .

4.9. Let (X, d) be a metric space. Let $C \subset X$ be a compact subset of X . Let $\{C_n : n = 1, 2, 3, \dots\}$ be a set of open subsets of X such that " $x \in C$ " implies " $x \in C_n$ for some n ." $\{C_n\}$ is called a countable open cover of C . Show that there is a finite integer N so that " $x \in C$ " implies " $x \in C_n$ for some integer $n < N$."

Proof Assume that an integer N does not exist such that " $x \in C$ " implies " $x \in C_n$ for some $n < N$." Then for each N we can find

$$x_N \in C \setminus \bigcup_{n=1}^N C_n.$$

Since $\{x_N\}_{N=1}^\infty$ is in C it possesses a subsequence with a limit $y \in C$. Clearly y does not belong to any of the subsets C_n . Hence “ $y \in C$ ” does not imply “ $y \in C_n$ for some integer n .” We have a contradiction. This completes the proof.

The following even stronger statement is true. Let (X, d) be a metric space. Let $C \subset X$ be compact. Let $\{C_i : i \in I\}$ denote *any* collection of open sets such that whenever $x \in C$, it is true that $x \in C_i$ for some index $i \in I$. Then there is a *finite* sub-collection, say $\{C_1, C_2, \dots, C_n\}$ such that $C \subset \bigcup_{i=1}^n C_i$. The point is that the original collection of open sets need not even be countably infinite. A good discussion of compactness in metric spaces can be found in [Mendelson 1963], Chapter V.

4.10. Let $X = (0, 1) \cup \{2\}$. That is, X consists of an open interval in $\mathbb{R}$, together with an “isolated” point. Show that the subsets $(0, 1)$ and $\{2\}$ of $(X, \text{Euclidean})$ are open. Show that $(0, 1)$ is closed in X . Show that $\{2\}$ is closed in X . Show that $\{2\}$ is compact in X but $(0, 1)$ is not compact in X .

Definition 4.5 Let $S \subset X$ be a subset of a metric space (X, d) . A point $x \in X$ is a boundary point of S if for every number $\epsilon > 0$, $B(x, \epsilon)$ contains a point in $X \setminus S$ and a point in S . The set of all boundary points of S is called the boundary of S and is denoted ∂S .

Definition 4.6 Let $S \subset X$ be a subset of a metric space (X, d) . A point $x \in S$ is called an interior point of S if there is a number $\epsilon > 0$ such that $B(x, \epsilon) \subset S$. The set of interior points of S is called the interior of S and is denoted S^0 .

Examples & Exercises

4.11. Let S be a subset of a metric space (X, d) . Show that $\partial S = \partial(X \setminus S)$. Deduce that $\partial X = \emptyset$.

4.12. Show that the property of being a boundary of a set is invariant under metric equivalence.

4.13. Let (X, d) be the real line with the Euclidean metric. Let S denote the set of all rational points in X (i.e., real numbers that can be written $\frac{p}{q}$ where p and q are integers with $q \neq 0$). Show that $\partial S = X$.

4.14. Find the boundary of $\mathbb{C}$ viewed as a subset of $(\hat{\mathbb{C}}, \text{spherical metric})$.

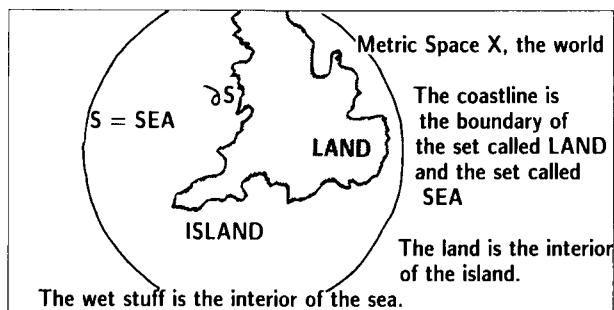
4.15. Let S be a closed subset of a metric space. Show that $\partial S \subset S$.

4.16. Let S be an open subset of a metric space. Show that $\partial S \cap S = \emptyset$.

4.17. Let S be an open subset of a metric space. Show that $S^0 = S$. Conversely, show that if $S^0 = S$ then S is open.

4.18. Let S be a closed subset of a metric space. Show that $S = S^0 \cup \partial S$.

Figure II.15. How well can topological concepts such as open, boundary, etc., be used to model land, sea, and coastlines?



4.19. Show that the property of being the interior of a set is invariant under metric equivalence.

4.20. Show that the boundary of a set S in a metric space always divides the space into two disjoint open sets whose union, with the boundary ∂S , is the whole space. Illustrate this result in the following cases, in the metric space $(\mathbb{R}^2, \text{Euclidean})$: (a) $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$; (b) $S = \mathbb{R}^2$.

4.21. Show that the boundary of a set is closed.

***4.22.** Let S be a subset of a compact metric space. Show that ∂S is compact.

4.23. Figure II.15 shows how we think of boundaries and interiors. What features of the picture are misleading?

4.24. To what extent does Mercator's projection provide a metric equivalence to a Cartesian map of the world?

4.25. Locate the boundary of the set of points marked in black in Figure II.16.

4.26. Prove the assertion made in the caption to Figure II.17.

5 Connected Sets, Disconnected Sets, and Pathwise-Connected Sets

Definition 5.1 A metric space (X, d) is connected if the only two subsets of X that are simultaneously open and closed are X and $\emptyset$. A subset $S \subset X$ is connected if the metric space (S, d) is connected. S is disconnected if it is not connected. S is totally disconnected provided that the only nonempty connected subsets of S are subsets consisting of single points.

Definition 5.2 Let $S \subset X$ be a subset of a metric space (X, d) . Then S is pathwise-connected if, for each pair of points x and y in S , there is a continuous function $f : [0, 1] \rightarrow S$, from the metric space $([0, 1], \text{Euclidean})$ into the metric space (S, d) , such that $f(0) = x$ and $f(1) = y$. Such a function f is called a path from x to y in S . S is pathwise-disconnected if it is not pathwise-connected.

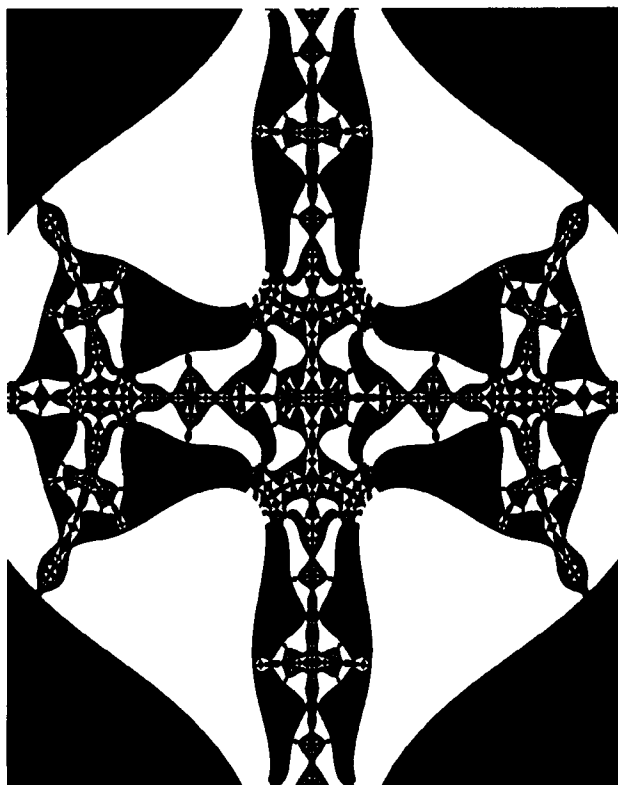


Figure II.16. Should the black part be called open and the white part closed? Locate the boundary of the set of points marked in black.

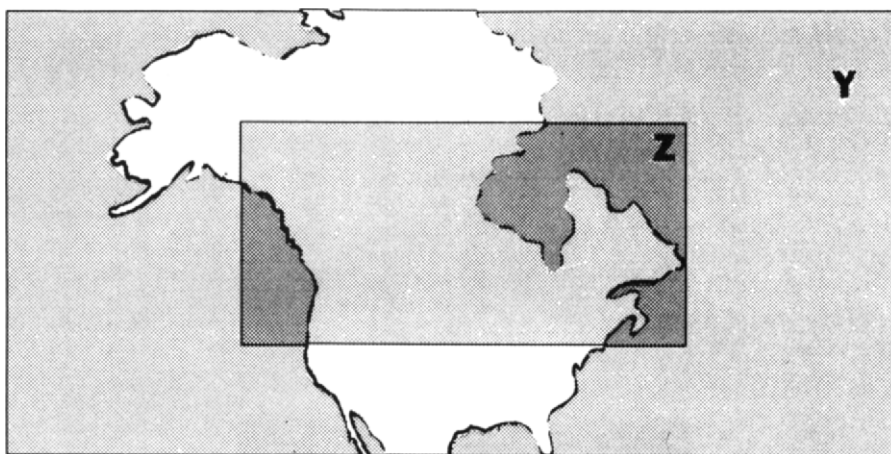


Figure II.17. The interior of the “land” set is an open set in the metric space $(Y = \text{land and water}, \text{Euclidean})$. The smaller filled rectangle denotes a subset $Z = \text{rectangle}$ of Y . The intersection of the interior of the land with Z is an open set in the metric space $(Z, \text{Euclidean})$, despite the fact that it includes some points of the “border” of Z .

One can also define *simply connected* and *multiply connected*. Let S be pathwise-connected. A pair of points $x, y \in S$ is *simply connected* in S if, given any two paths f_0 and f_1 connecting x, y in S , we can continuously deform f_0 to f_1 without leaving the subset S . What does this mean?

Let there be the two points $x, y \in S$ and the two paths f_0, f_1 connecting x, y in S . In other words, f_0, f_1 are two continuous functions mapping the unit interval $[0, 1]$ into S so that $f_0(0) = f_1(0) = x$ and $f_0(1) = f_1(1) = y$. By a *continuous deformation* of f_0 into f_1 within S we mean a function g continuously mapping the Cartesian product $[0, 1] \times [0, 1]$ into S , so that

$$(a) \quad g(s, 0) = f_0(s) (0 \leq s \leq 1)$$

$$(b) \quad g(s, 1) = f_1(s) (0 \leq s \leq 1)$$

$$(c) \quad g(0, t) = x (0 \leq t \leq 1)$$

$$(d) \quad g(1, t) = y (0 \leq t \leq 1)$$

Thus, we say that two points x, y in S are *simply connected* in S if, given any two paths f_0, f_1 going from x to y in S , there exists a function g as just described. This idea is illustrated in Figure II.18.

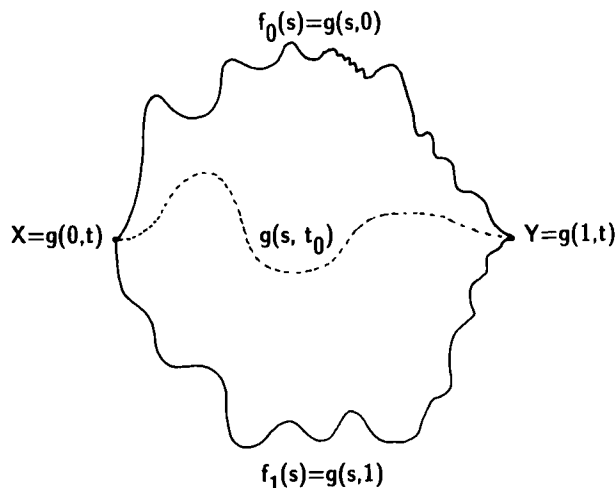
If x, y are not simply connected in S , then we say that x, y are *multiply connected* in S . S itself is called simply connected if every pair of points x, y in S is simply connected in S . Otherwise, S is called multiply connected. In the latter case we can imagine that S contains a “hole,” as illustrated in Figure II.19.

Examples & Exercises

***5.1.** Show that the properties of being (pathwise) connected, disconnected, simply connected, and multiply connected are invariant under metric equivalence.

5.2. Show that the metric space ($\mathbb{R}^n$, Euclidean) is simply connected.

Figure II.18. A path f_0 which connects the points x and y is continuously deformed, while remaining “attached” to x and y , to become a second path f_1 .



- 5.3.** Show that the metric space $(X = (0, 1) \cup \{2\}, \text{Euclidean})$ is disconnected.
- 5.4.** Show that the metric space $(\Sigma, \text{code space metric})$ is totally disconnected.
- *5.5.** Show that the metric space $(\bigcirc, \text{Manhattan})$ is multiply connected.
- 5.6.** Suppose $S_1 \supset S_2 \supset \cdots \supset S_n \supset \cdots$ is a nested sequence of nonempty connected subsets. Is $\bigcap_{n=1}^{\infty} S_n$ necessarily connected?
- 5.7.** Identify pathwise connected subsets of the metric space suggested in Figure II.20.
- 5.8.** Is $(\text{stick figure}, \text{Euclidean})$ simply or multiply connected?
- 5.9.** Discuss which set-theoretic properties (open, closed, connected, compact, bounded, etc.) would be best suited for a model of a cloud, treated as a subset of $\mathbb{R}^3$.
- 5.10.** The property that $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence in the metric space (X, d) is not invariant under homeomorphism but is invariant under metric equivalence, as illustrated in Figure II.21.

6 The Metric Space $(\mathcal{H}(X), h)$: The Space Where Fractals Live

We come to the ideal space in which to study fractal geometry. To start with, and always at the deepest level, we work in some complete metric space such as $(\mathbb{R}^2, \text{Euclidean})$ or $(\hat{\mathbb{C}}, \text{spherical})$, which we denote by (X, d) . But then, when we wish to discuss pictures, drawings, “black-on-white” subsets of the space, it becomes natural to introduce the space $\mathcal{H}$.

Definition 6.1 *Let (X, d) be a complete metric space. Then $\mathcal{H}(X)$ denotes the space whose points are the compact subsets of X , other than the empty set.*

Examples & Exercises

- 6.1.** Show that if x and $y \in \mathcal{H}(X)$ then $x \cup y$ is in $\mathcal{H}(X)$. Show that $x \cap y$ need not be in $\mathcal{H}(X)$. A picture of this situation is given in Figure II.22.
- 6.2.** What is the difference between a subset of $\mathcal{H}(X)$ and a compact nonempty subset of X ?

Definition 6.2 *Let (X, d) be a complete metric space, $x \in X$, and $B \in \mathcal{H}(X)$. Define*

$$d(x, B) = \min\{d(x, y) : y \in B\}.$$

$d(x, B)$ is called the distance from the point x to the set B .

Figure II.19. In a multiply connected space there exist paths that cannot be continuously deformed from one to another. There is some kind of “hole” in the space.

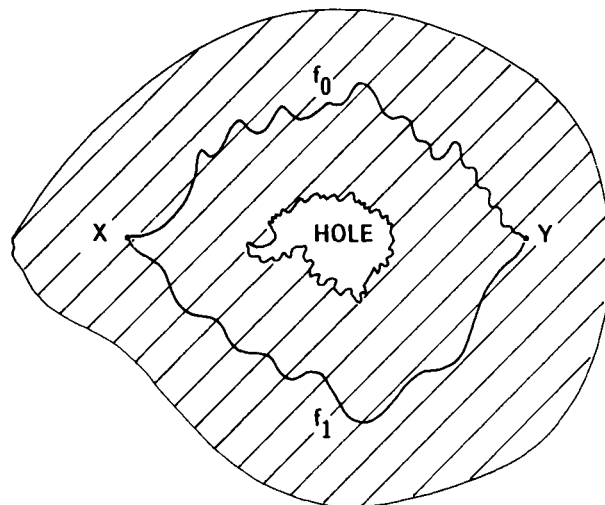
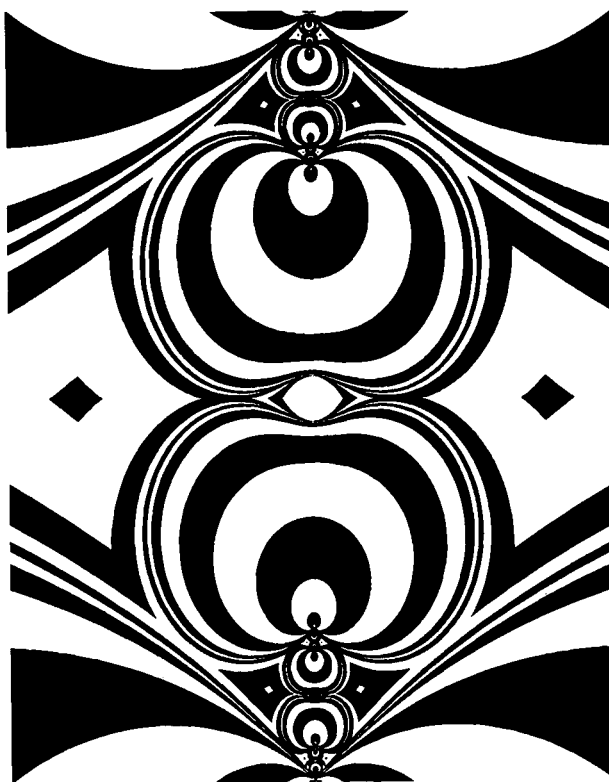


Figure II.20. Locate the largest connected subsets of this subset of $\mathbb{R}^2$.



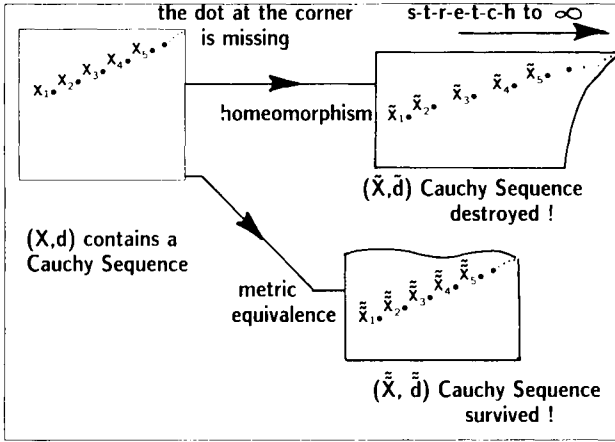


Figure II.21. A Cauchy sequence being preserved by a metric equivalence and destroyed by a certain homeomorphism.

How do we know that the set of real numbers $\{d(x, y) : y \in B\}$ contains a minimum value, as claimed in the definition? This follows from the compactness and nonemptiness of the set $B \in \mathcal{H}(X)$. Consider the function $f : B \rightarrow \mathbb{R}$ defined by

$$f(y) = d(x, y) \quad \text{for all } y \in B.$$

From the definition of the metric it follows that f is continuous, viewed as a transformation from the metric space (B, d) to the metric space $(\mathbb{R}, \text{Euclidean})$. Let $P = \inf\{f(y) : y \in B\}$, where “inf” is defined in exercise 6.19, and also in Definition 6.2 in Chapter III. Since $f(y) \geq 0$ for all $y \in B$, it follows that P is finite. We claim there is a point $\hat{y} \in B$ such that $d(x, \hat{y}) = P$. We can find an infinite sequence of points $\{y_n : n = 1, 2, 3, \dots\} \subset B$ such that $f(y_n) - P < \frac{1}{n}$ for each positive integer n . Using the compactness of B , we find that $\{y_n : n = 1, 2, 3, \dots\}$ has a limit $\hat{y} \in B$. Using the continuity of f we discover that $f(\hat{y}) = P$, which is what we needed to show.

Color Plate 2 shows a picture of the metric space ($\blacksquare$, Manhattan). It has been colored as follows. Let $\mathcal{F}$ denote a certain subset of $\blacksquare$ whose “geometry” is that of a piece of a fern. Then the color of each point $a \in \blacksquare$ is fixed by the value of $d(a, \mathcal{F})$.

Definition 6.3 Let (X, d) be a complete metric space. Let $A, B \in \mathcal{H}(X)$. Define

$$d(A, B) = \max\{d(x, B) : x \in A\}.$$

$d(A, B)$ is called the distance from the set $A \in \mathcal{H}(X)$ to the set $B \in \mathcal{H}(X)$.

Just as above, using the compactness of A and B , we can prove that this definition is meaningful. In particular, there are points $\hat{x} \in A$ and $\hat{y} \in B$ such that $d(A, B) = d(\hat{x}, \hat{y})$.

Examples & Exercises

6.3. Show that $B, C \in \mathcal{H}(X)$, with $B \subset C$, implies $d(x, C) \leq d(x, B)$.

6.4. Calculate $d(x, B)$ if (X, d) is the space $(\mathbb{R}^2, \text{Euclidean})$, $x \in \mathbb{R}^2$ is the point $(1, 1)$, and B is a closed disk of radius $\frac{1}{2}$ centered at the point $(\frac{1}{2}, 0)$.

6.5. Same as 6.4 above, but use the Manhattan metric.

6.6. Calculate $d(x, B)$ if (X, d) is $(\mathbb{R}, \text{Euclidean})$, $x = \frac{1}{2}$, and

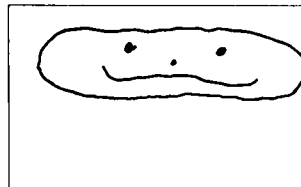
$$B = \{x_n = 3 + (-1)^n \frac{n}{n^2 + 1} : n = 1, 2, 3, \dots\} \cup \{3\}.$$

6.7. Let $A, B \in \mathcal{H}(X)$, where (X, d) is a metric space. Show that, in general, $d(A, B) \neq d(B, A)$. Conclude that d does not provide a metric on $\mathcal{H}(X)$. It is not symmetrical: the distance from A to B need not equal the distance from B to A .

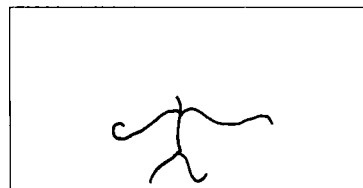
6.8. Figure II.23 shows two subsets A and B of $(\mathbb{R}^2, \text{Euclidean})$. A is the white part and B is the black part. (a) Estimate the location of a pair of points, $x \in A$ and $y \in B$, such that $d(x, y) = d(A, B)$. (b) Estimate the location of a pair of points, $x \in A$ and $y \in B$, such that $d(x, y) = d(B, A)$.

6.9. Figure II.24 shows two fern-like subsets, A and B , of $(\mathbb{R}^2, \text{Manhattan})$. Locate points $x \in A$ and $y \in B$ such that: (a) $d(x, y) = d(A, B)$; (b) $d(x, y) = d(B, A)$.

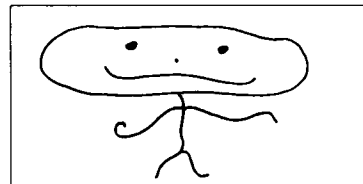
Figure II.22. Points in the space $\mathcal{H}(\mathbb{R}^2)$ may be interpreted as black-and-white images. Unions of points yield new points. Be careful with intersections, however.



The whole smiley face is a point in $\mathcal{H}(X)$. Call it $x \in \mathcal{H}(X)$.



This thin torso is a point in $\mathcal{H}(X)$. Call it $y \in \mathcal{H}(X)$.



This fellow is $x \cup y$. He is a single point in $\mathcal{H}(X)$.

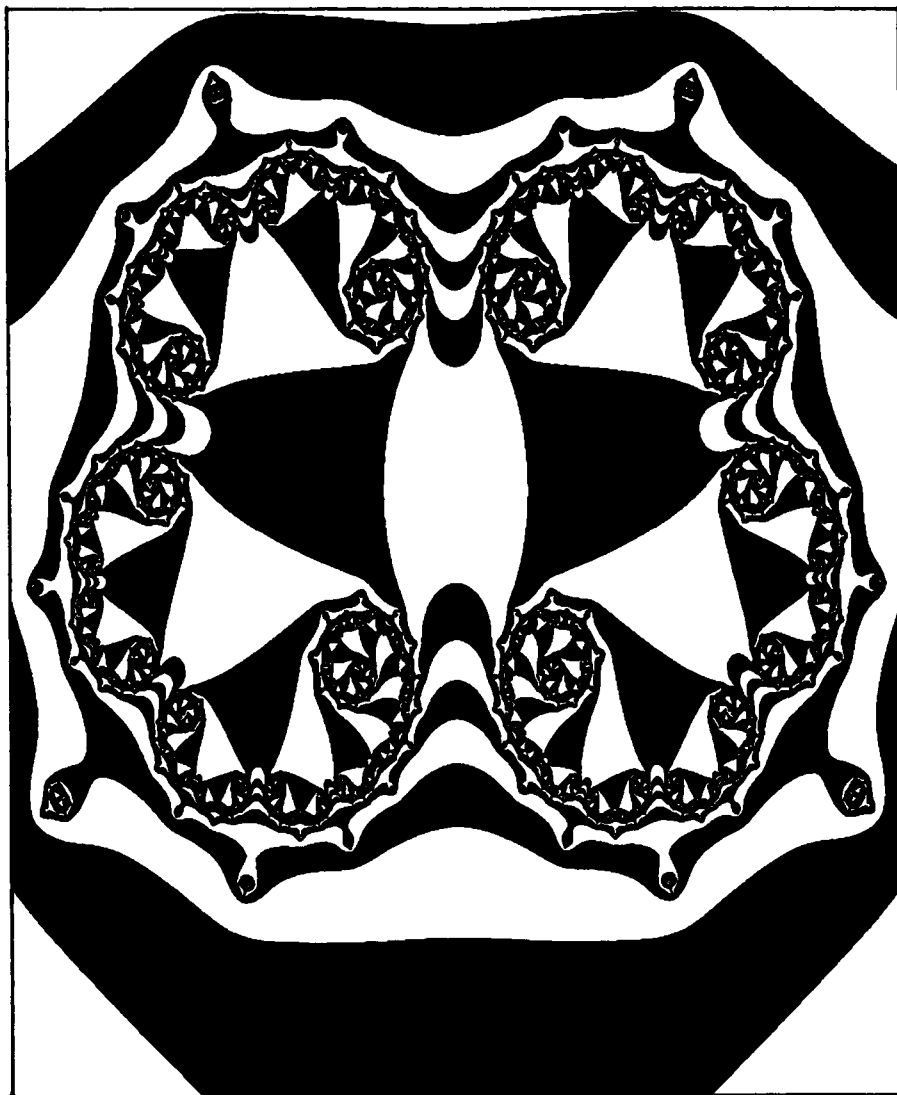


Figure II.23. This fractal image contains a pair of disjoint subsets of $\mathbb{R}^2$, “black” and “white.” Let A denote the closure of the set in black and let B denote the closure of its complement. Find a pair of points $x \in A$ and $y \in B$, such that $d(x, y) = d(A, B)$. Find a pair of points $\tilde{x} \in A$ and $\tilde{y} \in B$ such that $d(\tilde{x}, \tilde{y}) = d(B, A)$. Why do we “close” the sets before we begin?

6.10. Find $d(\text{France, U.S.A.})$ and $d(\text{U.S.A., France})$ on $(\hat{\mathbb{C}}, \text{spherical metric})$. Which is larger? Also compare $d(\text{Georgia, U.S.A.})$ to $d(\text{U.S.A., Georgia})$.

6.11. Let (X, d) be a complete metric space. Let A and B be points in $\mathcal{H}(X)$ such that $A \neq B$. Show that either $d(A, B) \neq 0$ or $d(B, A) \neq 0$. Show that if $A \subset B$ then $d(A, B) = 0$.

6.12. Let (X, d) be a complete metric space. Show that if A, B and $C \in \mathcal{H}(X)$ then $B \subset C \Rightarrow d(A, C) \leq d(A, B)$. (Hint: Use 6.3.)

6.13. Let (X, d) be a complete metric space. Show that if A, B , and $C \in \mathcal{H}(X)$ then

Figure II.24. Find a pair of points $\hat{x}$ and $\hat{y}$, one in the dark fern and one in the pale fern, such that the Hausdorff distance between the two fern images is the same as the distance between the points.



$$d(A \cup B, C) = d(A, C) \vee d(B, C).$$

We use the notation $x \vee y$ to mean the maximum of the two real numbers x and y .

Proof $d(A \cup B, C) = \max\{d(x, C) : x \in A \cup B\} = \max\{d(x, C) : x \in A\} \vee \max\{d(x, C) : x \in B\}.$

6.14. Let A , B , and C belong to $\mathcal{H}(X)$, where (X, d) is a metric space. Show that

$$d(A, B) \leq d(A, C) + d(C, B).$$

Also determine whether or not the inequality

$$d(A, B) \leq d(C, A) + d(C, B)$$

is true in general.

Definition 6.4 Let (X, d) be a complete metric space. Then the Hausdorff distance between points A and B in $\mathcal{H}(X)$ is defined by

$$h(A, B) = d(A, B) \vee d(B, A).$$

Examples & Exercises

6.15. Show that h is a metric on the space $\mathcal{H}(X)$.

Proof Let $A, B, C \in \mathcal{H}(X)$. Clearly $h(A, A) = d(A, A) \vee d(A, A) = d(A, A) = \max\{d(x, A) : x \in A\} = 0$. $h(A, B) = d(a, b)$ for some $a \in A$ and $b \in B$, using the compactness of A and B . Hence $0 \leq h(A, B) < \infty$. If $A \neq B$ we can assume there is an $a \in A$ so that $a \notin B$. It follows that $h(A, B) \geq d(A, B) > 0$. To show that $h(A, B) \leq h(A, C) + h(C, B)$ we first show that $d(A, B) \leq d(A, C) + d(C, B)$. We have for any $a \in A$

$$\begin{aligned} d(a, B) &= \min\{d(a, b) : b \in B\} \\ &\leq \min\{d(a, c) + d(c, b) : b \in B\} \forall c \in C, \\ &= d(a, c) + \min\{d(c, b) : b \in B\} \forall c \in C, \text{ so} \\ d(a, B) &\leq \min\{d(a, c) : c \in C\} + \max\{\min\{d(c, b) : b \in B\} : c \in C\}, \\ &= d(a, C) + d(C, B), \text{ so} \\ d(A, B) &= d(A, C) + d(C, B). \end{aligned}$$

Similarly,

$$\begin{aligned} d(B, A) &\leq d(B, C) + d(C, A), \text{ whence} \\ h(A, B) &= d(A, B) \vee d(B, A) \leq d(B, C) \vee d(C, B) + d(A, C) \vee d(C, A) \\ &= h(B, C) + h(A, C), \text{ as desired.} \end{aligned}$$

6.16. Show that $h(A \cup B, C \cup D) \leq h(A, C) \vee h(B, D)$, for all A, B, C , and $D \in \mathcal{H}(X)$.

6.17. Show that $h(A, B) = d(a, b)$ for some $a \in B$ and $b \in B$.

6.18. The same situation as in 6.9, but this time locate a pair of points $\hat{x} \in A$ and $\hat{y} \in B$ such that $d(\hat{x}, \hat{y}) = h(A, B)$, the Hausdorff distance from A to B .

6.19. Let $S \subset \mathbb{R}$, with $S \neq \emptyset$. The *supremum* of S is denoted by $\sup S$. The *infimum* of S is denoted by $\inf S$. If there is no real number greater than all of the numbers in S , then $\sup S = +\infty$; otherwise $\sup S = \min\{x \in \mathbb{R} : x \geq s \forall s \in S\}$. If there is no real number less than all of the numbers in S , then $\inf S = -\infty$; otherwise $\inf S = \max\{x \in \mathbb{R} : x \leq s \forall s \in S\}$. Show that $\sup S$ and $\inf S$ are well defined. Show that if S is compact then $\sup S = \max S$ and $\inf S = \min S$. Further exercises on $\sup$ and $\inf$ are given following Definition 6.2 in Chapter III.

By replacing $\max$ by $\sup$ and $\min$ by $\inf$, respectively, throughout the definition of the Hausdorff metric, define a “distance” between arbitrary pairs of subsets of a metric space. Give several reasons why this “distance” is not usually a metric.

7 The Completeness of the Space of Fractals

We refer to $(\mathcal{H}(X), h)$ as the “space of fractals.” It is too soon to be formal about the exact meaning of a “fractal.” At the present stage of development of science and mathematics, the idea of a fractal is most useful as a broad concept. Fractals are not defined by a short legalistic statement, but by the many pictures and contexts that

refer to them. For the first eight chapters of this book, any subset of $(\mathcal{H}(\mathbf{X}), h)$ is a fractal. However, as with the concept of a “space,” more meaning is suggested than is formalized.

In this section our principal goal is to establish that the space of fractals $(\mathcal{H}(\mathbf{X}), h)$ is a complete metric space. We also want to characterize convergent sequences in $\mathcal{H}(\mathbf{X})$. To achieve these goals using only the tools introduced so far is quite difficult. Indeed, at this juncture, we want to introduce another notion; namely, the idea of *extending* certain Cauchy subsequences.

Definition 7.1 Let $S \subset \mathbf{X}$ and let $\Gamma \geq 0$. Then $S + \Gamma = \{y \in \mathbf{X} : d(x, y) \leq \Gamma \text{ for some } x \in S\}$. $S + \Gamma$ is sometimes called, for example, in the theory of set morphology, the *dilation* of S by a ball of radius Γ .

Lemma 7.1 Let A and B belong to $\mathcal{H}(\mathbf{X})$ where $(\mathbf{X}, d)$ is a metric space. Let $\epsilon > 0$. Then

$$h(A, B) \leq \epsilon \Leftrightarrow A \subset B + \epsilon \quad \text{and} \quad B \subset A + \epsilon.$$

Proof Begin by showing that $d(A, B) \leq \epsilon \Leftrightarrow A \subset B + \epsilon$. Suppose $d(A, B) \leq \epsilon$. Then $\max\{d(a, B) : a \in A\} \leq \epsilon$ implies $d(a, B) \leq \epsilon$ for all $a \in A$. Hence for each $a \in A$ we have $a \in B + \epsilon$. Hence “ $A \subset B + \epsilon$.” Suppose “ $A \subset B + \epsilon$.” Consider $d(A, B) = \max\{d(a, B) : a \in A\}$. Let $a \in A$. Since $A \subset B + \epsilon$, there is $a, b \in B$ so that $d(a, b) \leq \epsilon$ for all $a \in A$. Hence $d(a, B) \leq \epsilon$. This is true for each $a \in A$. So “ $d(A, B) \leq \epsilon$.” This completes the proof.

Let $\{A_n : n = 1, 2, \dots, \infty\}$ be a Cauchy sequence of sets in $(\mathcal{H}(\mathbf{X}), h)$. That is, given $\epsilon > 0$, there is N so that $n, m \geq N$ implies

$$A_n + \epsilon \supset A_m \text{ and } A_m + \epsilon \supset A_n,$$

i.e., $h(A_n, A_m) \leq \epsilon$. We are concerned with Cauchy sequences $\{x_n\}_{n=1}^{\infty}$ in $\mathbf{X}$ with the property that $x_n \in A_n$ for each n . In particular, we need the following property that allows the *extension* of a Cauchy subsequence $\{x_{n_j} \in A_{n_j}\}_{j=1}^{\infty}$, with the property that $x_{n_j} \in A_{n_j}$ for each j , to a Cauchy sequence $\{x_n \in A_n\}_{n=1}^{\infty}$.

Lemma 7.2 The Extension Lemma. Let $(\mathbf{X}, d)$ be a metric space. Let $\{A_n : n = 1, 2, \dots, \infty\}$ be a Cauchy sequence of points in $(\mathcal{H}(\mathbf{X}), h)$. Let $\{n_j\}_{j=1}^{\infty}$ be an infinite sequence of integers

$$0 < n_1 < n_2 < n_3 < \dots$$

Suppose that we have a Cauchy sequence $\{x_{n_j} \in A_{n_j} : j = 1, 2, 3, \dots\}$ in $(\mathbf{X}, d)$. Then there is a Cauchy sequence $\{\tilde{x}_n \in A_n : n = 1, 2, \dots\}$ such that $\tilde{x}_{n_j} = x_{n_j}$, for all $j = 1, 2, 3, \dots$

Proof We give the construction of the sequence $\{\tilde{x}_n \in A_n : n = 1, 2, \dots\}$. For each $n \in \{1, 2, \dots, n_1\}$, choose $\tilde{x}_n \in \{x \in A_n : d(x, x_{n_1}) = d(x_{n_1}, A_n)\}$. That is, $\tilde{x}_n$ is the closest point (or one of the closest points) in A_n to x_{n_1} . The existence of such a

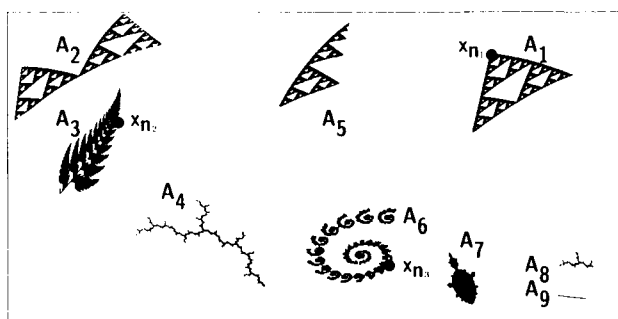


Figure II.25. The beginning of a Cauchy sequence $\{A_n\}$ of sets in $\mathcal{H}(\mathbb{R}^2)$ is shown. A Cauchy subsequence of points $\{x_{n_j}\}$ belonging to a subsequence of the sets is also indicated. Make a photocopy of the figure, and mark on it the extension of the subsequence of points to the visible sets in $\{A_n\}$.

closest point is ensured by the compactness of A_n . Similarly, for each $j \in \{2, 3, \dots\}$ and each $n \in \{n_j + 1, \dots, n_{j+1}\}$, choose $\tilde{x}_n \in \{x \in A_n : d(x, x_{n_j}) = d(x_{n_j}, A_n)\}$.

Now we show that $\{\tilde{x}_n\}$ has the desired properties, that it is indeed an extension of $\{x_{n_j}\}$ to $\{A_n\}$. Clearly $\tilde{x}_{n_j} = x_{n_j}$ and $x_n \in A_n$, by construction. To show that it is a Cauchy sequence let $\epsilon > 0$ be given. There is N_1 so that $n_k, n_j \geq N_1$ implies $d(x_{n_k}, x_{n_j}) \leq \epsilon/3$. There is N_2 so that $m, n \geq N_2$ implies

$$d(A_m, A_n) \leq \epsilon/3.$$

Let $N = \max\{N_1, N_2\}$ and note that, for $m, n \geq N$,

$$d(\tilde{x}_m, \tilde{x}_n) \leq d(\tilde{x}_m, x_{n_j}) + d(x_{n_j}, x_{n_k}) + d(x_{n_k}, \tilde{x}_n),$$

where $m \in \{n_{j-1} + 1, n_{j-1} + 2, \dots, n_j\}$ and $n \in \{n_{k-1} + 1, n_{k-1} + 2, \dots, n_k\}$. Since $h(A_m, A_{n_j}) < \epsilon/3$ there exists $y \in A_m \cap (\{x_{n_j}\} + \epsilon/3)$ so that $d(\tilde{x}_m, x_{n_j}) \leq \epsilon/3$. Similarly $d(x_{n_k}, \tilde{x}_n) \leq \epsilon/3$. Hence $d(\tilde{x}_m, \tilde{x}_n) \leq \epsilon$ for all $m, n > N$. This completes the proof.

Examples & Exercises

7.1. A Cauchy sequence $\{A_n\}$ of sets in $(\mathcal{H}(\mathbb{R}^2), h)$ is sketched in Figure II.25. The underlying metric space is $(\mathbb{R}^2, \text{Euclidean})$. A Cauchy subsequence $\{x_{n_j} \in A_{n_j}\}$ is also shown. Sketch, in the same figure, the extension $\{\tilde{x}_n\}$ of this subsequence to $\{A_n\}$.

7.2. Repeat 7.1 but this time with reference to Figure II.26.

The central result toward which we have been driving is this:

Theorem 7.1 The Completeness of the Space of Fractals. *Let (X, d) be a complete metric space. Then $(\mathcal{H}(X), h)$ is a complete metric space. Moreover, if $\{A_n \in \mathcal{H}(X)\}_{n=1}^\infty$ is a Cauchy sequence, then*

$$A = \lim_{n \rightarrow \infty} A_n \in \mathcal{H}(X)$$

can be characterized as follows:

$A = \{x \in X : \text{there is a Cauchy sequence } \{x_n \in A_n\} \text{ that converges to } x\}.$

Proof Let $\{A_n\}$ be a Cauchy sequence in $\mathcal{H}(X)$ and let A be defined as in the statement of the theorem. We break the proof up into the following parts:

- (a) $A \neq \emptyset$;
- (b) A is closed and hence complete since X is complete;
- (c) for $\epsilon > 0$ there is N such that for $n \geq N$, $A \subset A_n + \epsilon$;
- (d) A is totally bounded and thus by (b) is compact;
- (e) $\lim A_n = A$.

Proof of (a): We shall prove this part by proving the existence of a Cauchy sequence $\{a_i \in A_i\}$ in X . Toward this end find a sequence of positive integers $N_1 < N_2 < N_3 < \dots < N_n < \dots$ such that

$$h(A_m, A_n) < \frac{1}{2^i} \text{ for } m, n > N_i.$$

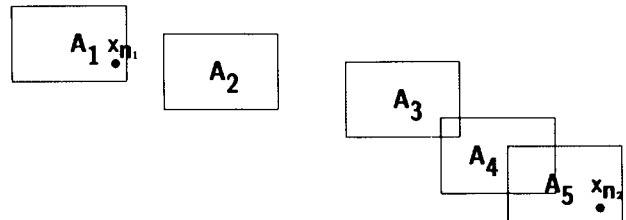
Choose $x_{N_1} \in A_{N_1}$. Then since $h(A_{N_1}, A_{N_2}) \leq \frac{1}{2}$, we can find an $x_{N_2} \in A_{N_2}$ such that $d(x_{N_1}, x_{N_2}) \leq \frac{1}{2}$. Assume that we have selected a finite sequence $x_{N_i} \in A_{N_i}$; $i = 1, 2, \dots, k$ for which $d(x_{N_{i-1}}, x_{N_i}) \leq \frac{1}{2^{i-1}}$. Then since $h(A_{N_k}, A_{N_{k+1}}) \leq \frac{1}{2^k}$, and $x_{N_k} \in A_{N_k}$, we can find $x_{N_{k+1}} \in A_{N_{k+1}}$ such that $d(x_{N_k}, x_{N_{k+1}}) \leq \frac{1}{2^k}$. For example let $x_{N_{k+1}}$ be the point in $A_{N_{k+1}}$ that is closest to x_{N_k} . By induction we can find an infinite sequence $\{x_{N_i} \in A_{N_i}\}$ such that $d(x_{N_i}, x_{N_{i+1}}) \leq \frac{1}{2^i}$. To see that $\{x_{N_i}\}$ is a Cauchy sequence in X , let $\epsilon > 0$ and choose N_ϵ such that $\sum_{i=N_\epsilon}^{\infty} \frac{1}{2^i} < \epsilon$. Then for $m > n \geq N_\epsilon$ we have

$$\begin{aligned} d(x_{N_m}, x_{N_n}) &\leq d(x_{N_m}, x_{N_{m+1}}) + d(x_{N_{m+1}}, x_{N_{m+2}}) + \dots + d(x_{N_{n-1}}, x_{N_n}) \\ &< \sum_{i=N_\epsilon}^{\infty} \frac{1}{2^i} < \epsilon. \end{aligned}$$

By the Extension Lemma, there exists a convergent subsequence $\{a_i \in A_i\}$ for which $a_{N_i} = x_{N_i}$. Then $\lim a_i$ exists and by definition is in A . Thus $A \neq \emptyset$.

Proof of (b): To show that A is closed, suppose $\{a_i \in A_i\}$ is a sequence that converges to a point a . We will show that $a \in A$, hence making A closed. For each positive integer i , there exists a sequence $\{x_{i,n} \in A_n\}$ such that $\lim_{n \rightarrow \infty} x_{i,n} = a_i$. There exists an increasing sequence of positive numbers $\{N_i\}_{i=1}^{\infty}$ such that $d(a_{N_i}, a) < \frac{1}{i}$.

Figure II.26. The same problem as for Figure II.25. The sets $\{A_n\}$ look very different here.



Furthermore, there is a sequence of integers $\{m_i\}$ such that $d(x_{N_i, m_i}, a_{N_i}) \leq \frac{1}{i}$. Thus $d(x_{N_i, m_i}, a) \leq 2/i$. If we let $y_{m_i} = x_{N_i, m_i}$ we see that $y_{m_i} \in A_{m_i}$ and $\lim i \rightarrow \infty y_{m_i} = a$. By the Extension Lemma, $\{y_{m_i}\}$ can be extended to a convergent sequence $\{z_i \in A_i\}$, and so $a \in A$. Thus we have shown A is closed.

Proof of (c): Let $\epsilon > 0$. There exists an N such that for $m, n \geq N$, $h(A_m, A_n) \leq \epsilon$. Now let $n \geq N$. Then for $m \geq n$, $A_m \subset A_n + \epsilon$. We need to show that $A \subset A_n + \epsilon$. To do this, let $a \in A$. There is a sequence $\{a_i \in A_i\}$ that converges to a . We may assume N is also large enough so that for $m \geq N$, $d(a_m, a) < \epsilon$. Then $a_m \in A_n + \epsilon$ since $A_m \subset A_n + \epsilon$. Since A_n is compact, it can be shown that $A_n + \epsilon$ is closed. Then since $a_m \in A_n + \epsilon$ for all $m \geq N$, a must also be in $A_n + \epsilon$. This completes the proof that $A \subset A_n + \epsilon$ for n large enough.

Proof of (d): Suppose A were not totally bounded. Then for some $\epsilon > 0$ there would not exist a finite ϵ -net. We could then find a sequence $\{x_i\}_{i=1}^\infty$ in A such that $d(x_i, x_j) \geq \epsilon$ for $i \neq j$. We shall show that this gives a contradiction. By (c) there exists an n large enough so that $A \subset A_n + \frac{\epsilon}{3}$. For each x_i , there is a corresponding $y_i \in A_n$ for which $d(x_i, y_i) \leq \frac{\epsilon}{3}$. Since A_n is compact, some subsequence $\{y_{n_i}\}$ of $\{y_i\}$ converges. Then we can find points in the sequence $\{y_{n_i}\}$ as close together as we wish. In particular we can find two points y_{n_i} and y_{n_j} such that $d(y_{n_i}, y_{n_j}) < \frac{\epsilon}{3}$. But then

$$d(x_{n_i}, x_{n_j}) \leq d(y_{n_i}, y_{n_i}) + d(y_{n_i}, y_{n_j}) + d(y_{n_j}, x_{n_j}) < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3},$$

and we have a contradiction to the way $\{x_{n_i}\}$ was chosen. Thus A is totally bounded and by part (b) compact.

Proof of (e): From part (d), $A \in \mathcal{H}(\mathbf{X})$. Hence by part (c) and Lemma 7.1 the proof that $\lim A_i = A$ will be complete if we show that for $\epsilon > 0$, there exists an N such that for $n \geq N$, $A_n \subset A + \epsilon$. To show this let $\epsilon > 0$ and find N such that for $m, n \geq N$, $h(A_m, A_n) \leq \frac{\epsilon}{2}$. Then for $m, n \geq N$, $A_m \subset A_n + \frac{\epsilon}{2}$. Let $n \geq N$. We will show that $A_n \subset A + \epsilon$. Let $y \in A_n$. There exists an increasing sequence $\{N_i\}$ of integers such that $n < N_1 < N_2 < N_3 < \dots < N_k < \dots$ and for $m, n \geq N_j$, $A_m \subset A_n + \frac{\epsilon}{2^{j+1}}$. Note that $A_n \subset A_{N_1} + \frac{\epsilon}{2}$. Since $y \in A_n$, there is an $x_{N_1} \in A_{N_1}$ such that $d(y, x_{N_1}) \leq \frac{\epsilon}{2}$. Since $x_{N_1} \in A_{N_1}$, there is a point $x_{N_2} \in A_{N_2}$ such that $d(x_{N_1}, x_{N_2}) \leq \frac{\epsilon}{2^2}$. In a similar manner we can use induction to find a sequence $x_{N_1}, x_{N_2}, x_{N_3}, \dots$, such that $x_{N_j} \in A_{N_j}$ and $d(x_{N_j}, x_{N_{j+1}}) < \frac{\epsilon}{2^{j+1}}$. Using the triangle inequality a number of times we can show that

$$d(y, x_{N_j}) \leq \frac{\epsilon}{2} \quad \text{for all } j$$

and also that $\{x_{N_j}\}$ is a Cauchy sequence. From the way n was chosen, each $A_{N_j} \subset A_n + \frac{\epsilon}{2}$. $\{x_{N_j}\}$ converges to a point x and since $A_n + \frac{\epsilon}{2}$ is closed, $x \in A_n + \frac{\epsilon}{2}$ also. Moreover, $d(y, x_{N_j}) \leq \epsilon$ implies that $d(y, x) \leq \epsilon$. We have thus shown that $A_n \subset A + \epsilon$ for $n \geq N$. This completes the proof that $\lim A_n = A$ and consequently that $(\mathcal{H}(\mathbf{X}), h)$ is a complete metric space.

Examples & Exercises

7.3. A tree waves in the wind. A special camera photographs the tree at times $t_n = (l - \frac{1}{n})$ sec, $n = 1, 2, 3, \dots$. Show that with reasonable assumptions the sequence of pictures thus obtained form a Cauchy sequence $\{A_n\}_{n=1}^{\infty}$ in $\mathcal{H}(\mathbb{R}^2)$. What does $A = \lim_{n \rightarrow \infty} A_n$ look like?

7.4. The Sierpinski triangle Δ is a compact subset of $(\mathbb{R}^2, \text{Euclidean})$. Hence $(\Delta, \text{Euclidean})$ is a compact metric space. Given an example of an infinite set in $(\mathcal{H}(\Delta), h)$, demonstrate a Cauchy sequence $\{A_n \in \mathcal{H}(\Delta)\}$ that is contained in your set and describe its limit.

7.5. Figure II.27 shows a convergent sequence of sets in $\mathcal{H}(\blacksquare)$ converging to a fern. Pick a point in A . Find a Cauchy sequence $\{x_n \in A_n\}$ that converges to it.

***7.6.** Let (X, d) be a compact metric space. Show that $(\mathcal{H}(X), h)$ is a compact metric space, where h is the Hausdorff metric on the space $\mathcal{H}(X)$.

It's a good idea to get familiar with $\mathcal{H}(X)$, by trying to see what properties of the space X are also true of $\mathcal{H}(X)$. We have seen that completeness is one of these, and exercise 7.6 shows that compactness is as well. So are some forms of connectedness, and by way of example we prove this for path-connectedness of $\mathcal{H}(\mathbb{R})$ as follows:

Theorem 7.2 *The function $f : \mathbb{R} \rightarrow \mathcal{H}$ given by $f(x) = \{x\}$ is continuous.*

Proof Let $\{x_n\}$ be a sequence in $\mathbb{R}$ that converges to a point x . Then given $\epsilon > 0$ there is an N such that for $n > N$, $d(x_n, x) < \epsilon$. By definition, $h(\{x_n\}, \{x\}) = d(x_n, x)$, since there is only one element in each set. Hence $\{\{x_n\}\}$ converges to $\{x\}$ in $\mathcal{H}(\mathbb{R})$. So the function $f(x)$ is continuous. Thus the image of $\mathbb{R}$ in $\mathcal{H}(\mathbb{R})$ is path-connected. Notice that a function like this guarantees that there will be a copy of any space in its associated space of non-empty compact sets.

Theorem 7.3 *The functions $f_x : [0, 1] \rightarrow \mathcal{H}$ given by $f_x(a) = [x, x + a]$, $0 \leq a \leq 1$ are continuous, thereby showing that there is a path in $\mathcal{H}$ from an interval to one of its endpoints.*

Proof As before, let $\{a_n\}$ be a convergent sequence in $[0, 1]$ with limit a . Suppose $d(a_n, a) < \epsilon$. Then the distance $h([x, x + a_n], [x, x + a])$ is given by

$$\begin{aligned} h([x, x + a_n], [x, x + a]) &= d([x, x + a_n], [x, x + a]) \vee d([x, x + a], [x, x + a_n]) \\ &= d(a, a_n) < \epsilon. \end{aligned}$$

Hence each of the functions f_x is continuous.

Theorem 7.4 *If A is a compact subset of $\mathbb{R}$ then the function $f_A : [0, b] \rightarrow \mathcal{H}$ given by $f_A(a) = \bigcup [x, x + a]$ such that $x \in A$ is continuous.*

Proof By part (b), the function f_A is continuous at each point in A , when $b = 1$. Hence the statement is true for $b = 1$. To get the whole statement, we note that if $g : [0, b] \rightarrow [0, 1]$ is given by $g(x) = (1/b)x$, g is continuous. Hence f_A can be written as the composition of two continuous functions and is continuous.

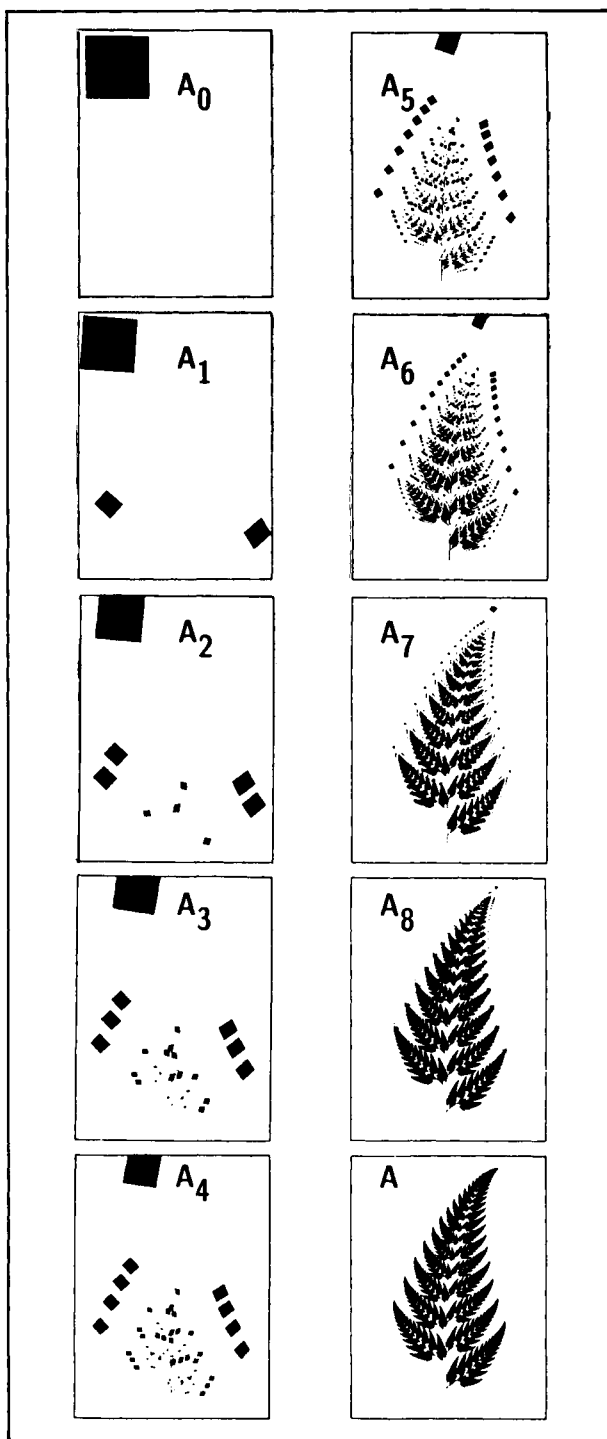


Figure 11.27. A Cauchy sequence of sets $\{A_n\}$ in the space $\mathcal{H}(\mathbb{R}^2)$ converging to a fern-like set.

Theorem 7.5 *If A is a compact subset of $\mathbb{R}$ then the set $\bigcup [x, x + b]$ such that $x \in A$ is an interval for b large enough.*

Proof A is compact, hence it is bounded. If we take b to be the length of any interval $[c, d]$ such that $A \subset [c, d]$, then the set $[x, x + b]$ with x the least element of A will overlap the set $[y, y + b]$ where y is the greatest element of A . Thus all the sets of this form will overlap, and their union forms one interval, which is path-connected.

Theorem 7.6 *If A and B are compact subsets of $\mathbb{R}$ then there is a path in $\mathcal{H}$ connecting them.*

Proof For any space X and three points $a, b, c \in X$, if there is a path from a to b and one from b to c then there is one from a to c . We construct a path between A and B in $\mathcal{H}(\mathbb{R})$ by taking first a path from A to the interval formed in part (d), namely choose b as in (d) and construct the path

$$f : [0, 1] \rightarrow [0, b] \rightarrow \{f_A(x) : 0 \leq x \leq b\},$$

which is continuous by part (c). Thus there is a path between every point in $\mathcal{H}(\mathbb{R})$ and a point that is an interval. Similarly, there is a path from some interval to every point in $\mathcal{H}(\mathbb{R})$. By part (b) there is a path from every interval to one of its endpoints. By part (a) there is a path (the image of $\mathbb{R}$) between any two of these endpoints. Therefore a path from A to B may be constructed by taking A to an interval, taking the interval to an endpoint, moving to the a new point of $\mathbb{R}$, making a path to an interval, and taking the path from the interval to B . Hence $\mathcal{H}(\mathbb{R})$ is pathwise-connected. It is true, though somewhat more complicated, that if X is connected then so is $\mathcal{H}(X)$.

8 Additional Theorems about Metric Spaces

We state here a number of theorems that we shall use later on. Full proofs are not provided. They can be found in most introductory topology texts. We particularly recommend [Kasriel 1971] and [Mendelson 1963]. These theorems may be treated as exercises in metric space theory.

Theorem 8.1 *Let (X, d) be a metric space. Let $\{x_n\}$ be a Cauchy sequence convergent to $x \in X$ (or equivalently let $\{x_n\}$ be a sequence and x be a point, such that $\lim_{n \rightarrow \infty} d(x, x_n) = 0$). Let $f : X \rightarrow X$ be continuous. Then*

$$\lim_{n \rightarrow \infty} f(x_n) = f(x).$$

Proof See your first calculus book.

Theorem 8.2 *Let (X_1, d_1) and (X_2, d_2) be metric spaces. Let $f : X_1 \rightarrow X_2$ be continuous. Let $E \subset X_1$ be compact. Then $f : E \rightarrow X_2$ is uniformly continuous: that*

is, given $\epsilon > 0$ there is a number $\delta > 0$ so that

$$d_2(f(x), f(y)) < \epsilon \text{ whenever } d_1(x, y) < \delta \text{ for all } x, y \in E.$$

Proof Use the fact that any open cover of E contains a finite subcover.

Theorem 8.3 Let (X_i, d_i) be metric spaces for $i = 1, 2, 3$. Let $f : X_1 \times X_2 \rightarrow X_3$ have the following property. For each $\epsilon > 0$ there exists $\delta > 0$ such that

$$(i) \ d_1(x_1, y_1) < \delta \Rightarrow d_3(f(x_1, x_2), f(y_1, y_2)) < \epsilon,$$

$$\forall x_1, y_1 \in X_1$$

$$\forall x_2 \in X_2$$

and

$$(ii) \ d_2(x_2, y_2) < \delta \Rightarrow d_3(f(y_1, x_2), f(y_1, y_2)) < \epsilon,$$

$$\forall y_1 \in X_1$$

$$\forall x_2, y_2 \in X_2$$

Then f is continuous on the metric space $(X = X_1 \times X_2, d)$, where $d((x_1, x_2), (y_1, y_2)) = \max\{d_1(x_1, y_1), d_2(x_2, y_2)\}$.

Proof Use

$$d(f(x_1, x_2), f(y_1, y_2)) \leq d(f(x_1, x_2), f(y_1, x_2)) + d(f(y_1, x_2), f(y_1, y_2)),$$

but check first that d is a metric.

Theorem 8.4 Let (X_i, d_i) be metric spaces for $i = 1, 2$ and let the metric space (X, d) be defined as in Theorem 8.3. If $K_1 \subset X_1$ and $K_2 \subset X_2$ are compact, then $K_1 \times K_2 \subset X$ is compact.

Proof Deal with the component in K_1 first.

Theorem 8.5 Let (X_i, d_i) be compact metric spaces for $i = 1, 2$. Let $f : X_1 \rightarrow X_2$ be continuous, one-to-one, and onto. Then f is a homeomorphism.